Cognitive APIs

APIs and Cognitive APIs

- **Cognitive APIs:** APIs that provide a cognitive (data science) service.
- **Multiple cloud providers provide such cognitive APIs:** Microsoft, Amazon, Google, IBM, Twitter, etc.
- Features of APIs
  - Input, Output
  - Rate limits
  - Authentication
    - HTTP auth: provide username and password.
    - API keys: a unique generated value is assigned to each first time user
    - OAuth
  - Pricing

Azure Cognitive Service APIs

Phi-3 open models

Build with the most capable and cost-effective small language models (SLMs) available.

Learn more

Azure OpenAI Service

Build your own copilot and generative AI applications with cutting-edge language and vision models.

Learn more

Azure AI Search

Retrieve the most relevant data using keyword, vector, and hybrid search.

Learn more

Azure AI Content Safety

Monitor text and images to detect offensive or inappropriate content.

Learn more

Azure AI Translator

Translate documents and text in real time across more than 100 languages.

Learn more

Azure AI Speech

Use industry-leading AI services such as speech-to-text, text-to-speech, speech translation, and speaker recognition.

Learn more

Azure AI Vision

Read text, analyze images, and detect faces with optical character recognition (OCR) and machine learning.

Learn more

Azure AI Language

Build conversational interfaces, summarize documents, and analyze text using prebuilt AI-powered features.

Learn more

Azure AI Document Intelligence

Apply advanced machine learning to extract text, key-value pairs, tables, and structures from documents.

Learn more

Azure AI Vision

Image analysis

Image analysis that pulls from more than 10,000 concepts and objects to detect, classify, caption, and generate insights.

Optical character recognition (OCR)

Optical character recognition (OCR) to extract printed and handwritten text from images with varied languages and writing styles.

Facial recognition

Facial recognition to create intelligent applications that recognize and verify human identity.

Optical Character Recognition

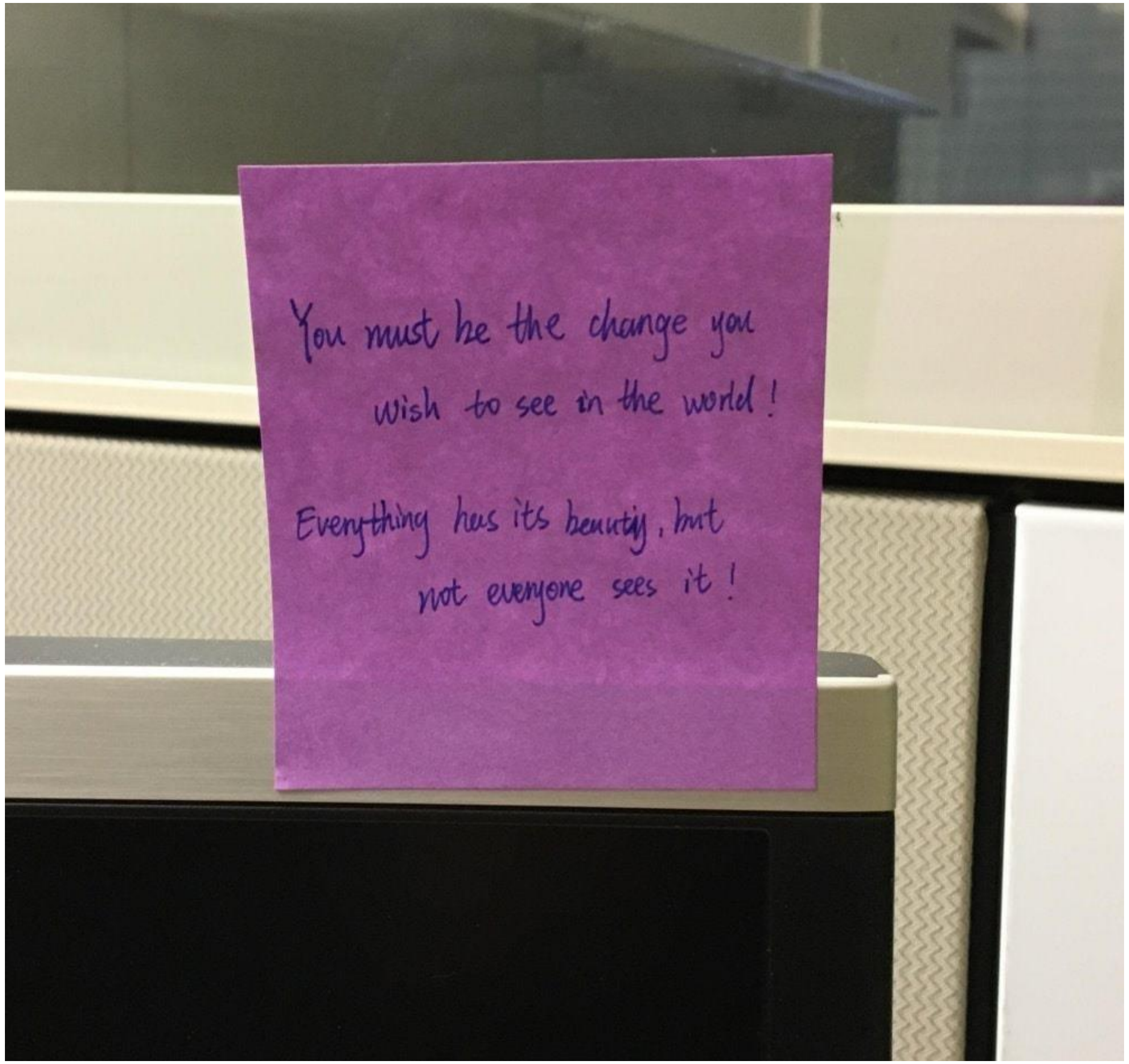
Use one of your own files or choose from a sample below.

Drag and drop a file here
or
Browse for a file
or
Take a photo

Sample form #3

Detected attributes JSON

Nutrition Facts Amount Per Serving
Serving size: 1 bar (40g)
Serving Per Package: 4
Total Fat 13g
Saturated Fat 1.5g
Amount Per Serving
Trans Fat 0g
alories 190
Cholesterol 0mg
ories from Fat 110
Sodium 20mg
nt Daily Values are based on
Vitamin A 50%
calorie diet.



NSE talent sprint IIIT Hyderabad

Read handwritten text from images

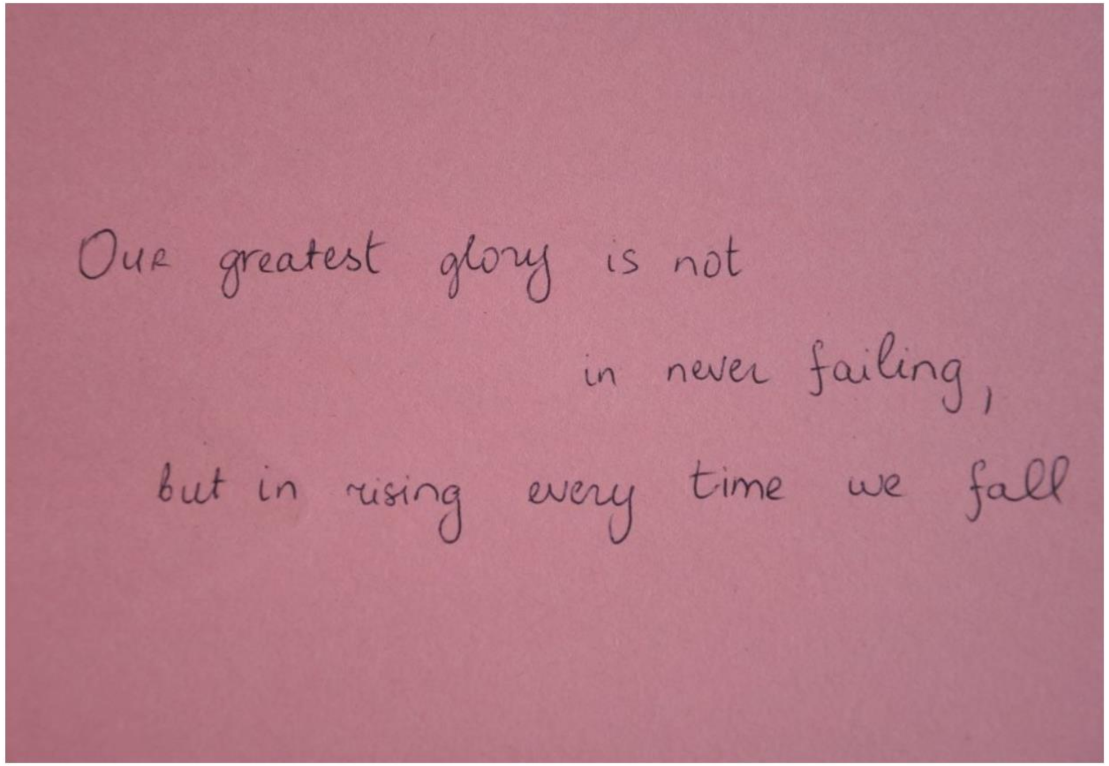


Image URL

Submit

Browse

Preview | JSON

OUR greatest glory is not

in never failing,

om

but in rising

_every

time

fall

People Detection


```
"peopleResult": {
  "values": [
    {
      "boundingBox": {
        "x": 0,
        "y": 41,
        "w": 95,
        "h": 189
      },
      "confidence": 0.9474349617958069
    },
    {
      "boundingBox": {
        "x": 204,
        "y": 96,
        "w": 95,
        "h": 134
      },
      "confidence": 0.9470965266227722
    },
```

```
{
  "boundingBox": {
    "x": 53,
    "y": 20,
    "w": 136,
    "h": 210
  },
  "confidence": 0.8943784832954407
},
{
  "boundingBox": {
    "x": 170,
    "y": 31,
    "w": 91,
    "h": 199
  },
  "confidence": 0.2713555097579956
}
```


Image Description


```
{
  "description":{
    "tags":[
      "outdoor",
      "city",
      "white"
    ],
    "captions":[
      {
        "text":"a city with tall buildings",
        "confidence":0.48468858003616333
      }
    ]
  },
  "requestId":"7e5e5cac-ef16-43ca-a0c4-02bd49d379e9",
  "metadata":{
    "height":300,
    "width":239,
    "format":"Png"
  },
  "modelVersion":"2021-05-01"
}
```

Object Detection

```
{
    "name": "kitchen appliance",
    "confidence": 0.501,
    "boundingBox": {"x":730,"y":66,"w":135,"h":85}
},
{
    "name": "computer keyboard",
    "confidence": 0.51,
    "boundingBox": {"x":523,"y":377,"w":185,"h":46}
},
{
    "name": "Laptop",
    "confidence": 0.85,
    "boundingBox": {"x":471,"y":218,"w":289,"h":226}
},
{
    "name": "person",
    "confidence": 0.855,
    "boundingBox": {"x":654,"y":0,"w":584,"h":473}
}
```

Image tagging

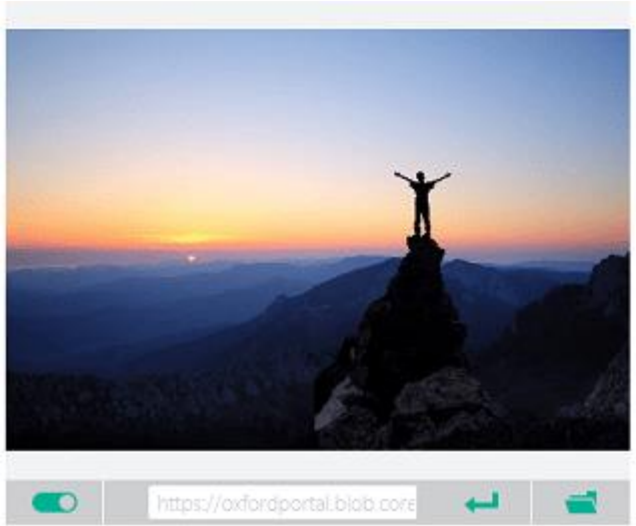
{"name": "grass","confidence": 0.996}, {"name": "outdoor","confidence": 0.996}, {"name": "building","confidence": 0.989}, {"name": "property","confidence": 0.985}, {"name": "plant","confidence": 0.979}, {"name": "sky","confidence": 0.976}, {"name": "home","confidence": 0.973}, {"name": "house","confidence": 0.973}, {"name": "real estate","confidence": 0.972}, {"name": "yard","confidence": 0.948}, {"name": "siding","confidence": 0.945}, {"name": "porch","confidence": 0.941}, {"name": "cottage","confidence": 0.914}, {"name": "tree","confidence": 0.911}, {"name": "farmhouse","confidence": 0.899}, {"name": "window","confidence": 0.895}, {"name": "lawn","confidence": 0.894}, {"name": "backyard","confidence": 0.893}, {"name": "garden buildings","confidence": 0.886}, {"name": "roof","confidence": 0.870}, {"name": "driveway","confidence": 0.867}, {"name": "land lot","confidence": 0.856}, {"name": "landscaping","confidence": 0.854}

Smart-cropped thumbnails

1. Remove distracting elements from the image and identify the *area of interest*—the area of the image in which the main object(s) appears.

2. Crop the image based on the identified *area of interest*.

3. Change the aspect ratio to fit the target thumbnail dimensions.



Brand detection


```
"brands":[
   {
      "name":"Microsoft",
      "rectangle":{
         "x":20,
         "y":97,
         "w":62,
         "h":52
      }
   }
]
```


Celebrity and Landmark detection


```
"celebrities": [{
  "faceRectangle": {
    "top": 391,
    "left": 318,
    "width": 184,
    "height": 184
  },
  "name": "Satya Nadella",
  "confidence": 0.99999856948852539
}]
```

Image categorization

86-category taxonomy

- abstract_
  - abstract_net
  - abstract_nonphoto
  - abstract_rect
  - abstract_shape
  - abstract_texture
- animal_
  - animal_bird
  - animal_cat
  - animal_dog
  - animal_horse
  - animal_panda
- building_
  - building_arch
  - building_brickwall
  - building_church
  - building_corner
  - building_doorwindows
  - building_pillar
  - building_stair
  - building_street
- dark_
  - dark_fire
  - dark_fireworks
  - dark_light
- drink_
  - drink_can
- food_
  - food_bread
  - food_fastfood
  - food_grilled
  - food_pizza
- indoor_
  - indoor_churchwindow
  - indoor_court
  - indoor_doorwindows
  - indoor_marketstore
  - indoor_room
  - indoor_venue
- object_screen
- object_sculpture
- others_
- outdoor_
  - outdoor_city
  - outdoor_field
  - outdoor_grass
  - outdoor_house
  - outdoor_mountain
  - outdoor_oceanbeach
  - outdoor_playground
  - outdoor_pool
  - outdoor_railway
  - outdoor_road
  - outdoor_sportsfield
  - outdoor_stonerock
  - outdoor_street
  - outdoor_water
  - outdoor_waterside
- people_
  - people_baby
  - people_crowd
  - people_group
  - people_hand
  - people_many
  - people_portrait
  - people_show
  - people_swimming
  - people_tattoo
  - people_young
- plant_
  - plant_branch
  - plant_flower
  - plant_leaves
  - plant_tree
- sky_cloud
- sky_object
- sky_sun
- text_
  - text_mag
  - text_map
  - text_menu
  - text_sign
- trans_bicycle
- trans_bus
- trans_car
- trans_trainstation

people_group

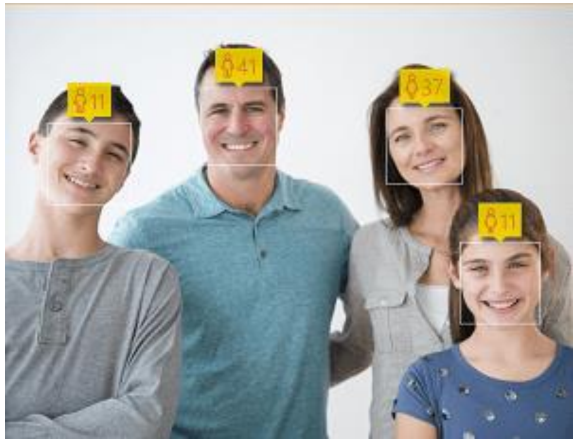
animal_dog

outdoor_mountain

food_bread

Face Detection


```
"faces": [
    {
        "age": 23,
        "gender": "Female",
        "faceRectangle": {
            "top": 45,
            "left": 194,
            "width": 44,
            "height": 44
        }
    }
],
```


```
"faces": [
    {
        "age": 11,
        "gender": "Male",
        "faceRectangle": {
            "top": 62,
            "left": 22,
            "width": 45,
            "height": 45
        }
    },
    {
        "age": 11,
        "gender": "Female",
        "faceRectangle": {
            "top": 127,
            "left": 240,
            "width": 42,
            "height": 42
        }
    },
```

```
{
    "age": 37,
    "gender": "Female",
    "faceRectangle": {
        "top": 55,
        "left": 200,
        "width": 41,
        "height": 41
    }
},
{
    "age": 41,
    "gender": "Male",
    "faceRectangle": {
        "top": 45,
        "left": 103,
        "width": 39,
        "height": 39
    }
}
```

Face Detection

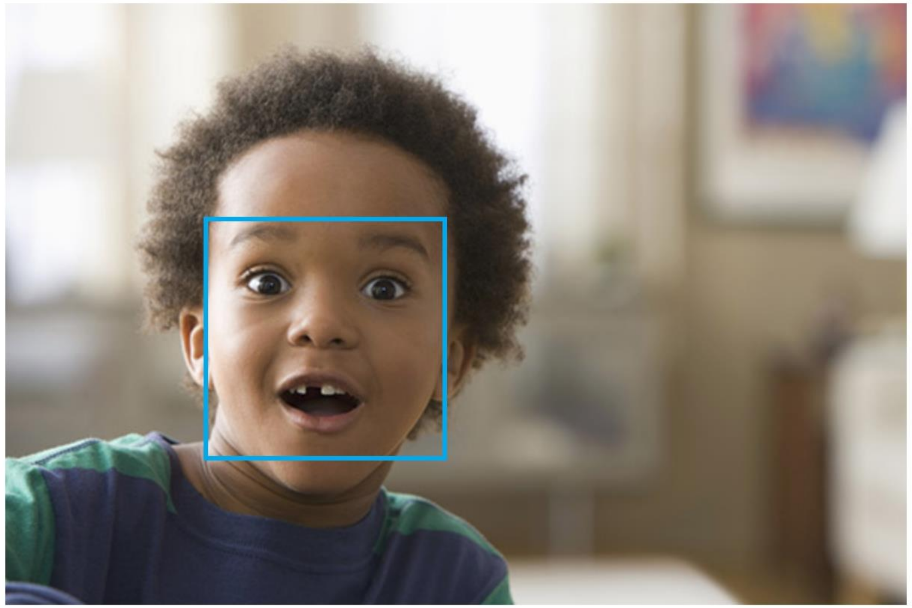
JSON: [{
"faceId": "add3ab06-6d7f-48ee-94c4-6d97abcfa34d",
"faceRectangle": { "top": 128, "left": 459, "width": 224, "height": 224 },
"faceAttributes": {
"hair": {
"bald": 0.1,
"invisible": false,
"hairColor": [{ "color": "brown", "confidence": 0.99 }, { "color": "black", "confidence": 0.57 }, { "color": "red", "confidence": 0.36 }, { "color": "blond", "confidence": 0.34 }, { "color": "gray", "confidence": 0.15 }, { "color": "other", "confidence": 0.13 }]
},
"smile": 1.0,
"headPose": { "pitch": 0.0, "roll": -16.9, "yaw": 16.7 },
"gender": "female",
"age": 24.0,
"facialHair": { "moustache": 0.0, "beard": 0.0, "sideburns": 0.0 },
"glasses": "ReadingGlasses",
"makeup": { "eyeMakeup": true, "lipMakeup": true },
"emotion": { "anger": 0.0, "contempt": 0.0, "disgust": 0.0, "fear": 0.0, "happiness": 1.0, "neutral": 0.0, "sadness": 0.0, "surprise": 0.0 },
"occlusion": { "foreheadOccluded": false, "eyeOccluded": false, "mouthOccluded": false },
"accessories": [{ "type": "glasses", "confidence": 1.0 }],
"blur": { "blurLevel": "low", "value": 0.0 },
"exposure": { "exposureLevel": "goodExposure", "value": 0.48 },
"noise": { "noiseLevel": "low", "value": 0.0 }
},
"faceLandmarks": {
"pupilLeft": { "x": 504.8, "y": 206.8 }, "pupilRight": { "x": 602.5, "y": 178.4 }, "noseTip": { "x": 593.5, "y": 247.3 }, "mouthLeft": { "x": 529.8, "y": 300.5 }, "mouthRight": { "x": 626.0, "y": 277.3 }, "eyebrowLeftOuter": { "x": 461.0, "y": 186.8 }, "eyebrowLeftInner": { "x": 541.9, "y": 178.9 }, "eyeLeftOuter": { "x": 490.9, "y": 209.0 }, "eyeLeftTop": { "x": 509.1, "y": 199.5 }, "eyeLeftBottom": { "x": 509.3, "y": 213.9 }, "eyeLeftInner": { "x": 529.0, "y": 205.0 }, "eyebrowRightInner": { "x": 579.2, "y": 169.2 }, "eyebrowRightOuter": { "x": 633.0, "y": 136.4 }, "eyeRightInner": { "x": 590.5, "y": 184.5 }, "eyeRightTop": { "x": 604.2, "y": 171.5 }, "eyeRightBottom": { "x": 608.4, "y": 184.0 }, "eyeRightOuter": { "x": 623.8, "y": 173.7 }, "noseRootLeft": { "x": 549.8, "y": 200.3 }, "noseRootRight": { "x": 580.7, "y": 192.3 }, "noseLeftAlarTop": { "x": 557.2, "y": 234.6 }, "noseRightAlarTop": { "x": 603.2, "y": 225.1 }, "noseLeftAlarOutTip": { "x": 545.4, "y": 255.5 }, "noseRightAlarOutTip": { "x": 615.9, "y": 239.5 }, "upperLipTop": { "x": 591.1, "y": 278.4 }, "upperLipBottom": { "x": 593.2, "y": 288.7 }, "underLipTop": { "x": 597.1, "y": 308.0 }, "underLipBottom": { "x": 600.3, "y": 324.8 } } }]

Emotion Recognition

Emotion recognition

The Face API now integrates emotion recognition, returning the confidence across a set of emotions for each face in the image such as anger, contempt, disgust, fear, happiness, neutral, sadness, and surprise. These emotions are understood to be cross-culturally and universally communicated with particular facial expressions.

See it in action



Detection result:
1 faces detected

JSON:


```
[
  {
    "faceRectangle": {
      "top": 141,
      "left": 130,
      "width": 162,
      "height": 162
    },
    "scores": {
      "anger": 9.29041E-06,
      "contempt": 0.000118981574,
      "disgust": 3.15619363E-05,
      "fear": 0.000589638,
      "happiness": 0.06630674,
      "neutral": 0.00555004273,
      "sadness": 7.44669524E-06,
      "surprise": 0.9273863
    }
  }
]
```


Image URL

Submit

Browse

Face Verification

Face verification

Check the likelihood that two faces belong to the same person. The API will return a confidence score about how likely it is that the two faces belong to one person.

See it in action

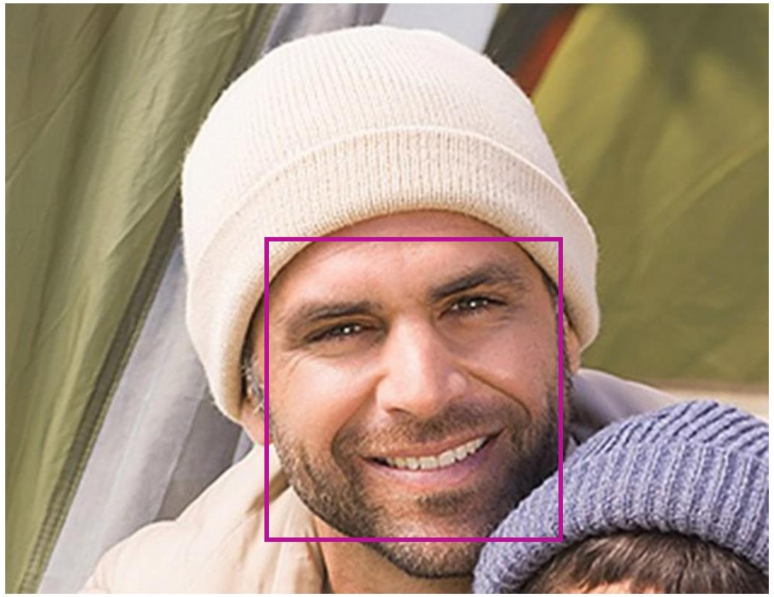


Image URL Submit Browse

Image URL Submit Browse

Verification result: The two faces belong to the same person. **Confidence is 0.7349.**

Image Type Detection

- Clip art type and line drawing detection

Value	Meaning
0	Non-clip-art
1	Ambiguous
2	Normal-clip-art
3	Good-clip-art

"clipArtType": 3,
"lineDrawingType": 0

"clipArtType": 0,
"lineDrawingType": 0

"clipArtType": 2,
"lineDrawingType": 1

"clipArtType": 0,
"lineDrawingType": 0

Color scheme detection

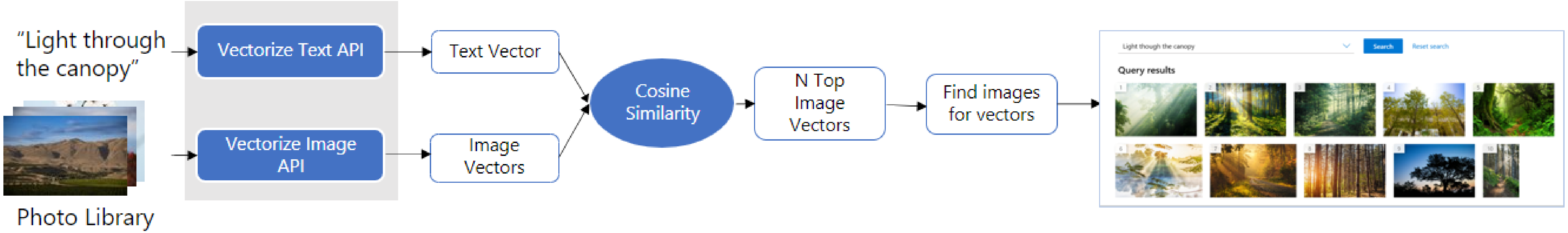

```
{
    "color": {
        "dominantColorForeground": "Black",
        "dominantColorBackground": "Black",
        "dominantColors": ["Black", "White"],
        "accentColor": "BB6D10",
        "isBwImg": false
    },
    "requestId": "0dc394bf-db50-4871-bdcc-13707d9405ea",
    "metadata": {
        "height": 202,
        "width": 300,
        "format": "Jpeg"
    }
}
```


Foreground: Black
Background: White
Colors: Black, White, Green

Foreground: Black
Background: Black
Colors: Black

Multimodal embeddings

- Generating a vector representation of an image that captures its features and characteristics.
- Useful for image retrieval.
- Multimodal embedding is not designed analyze medical images for diagnostic features or disease patterns.



Background removal

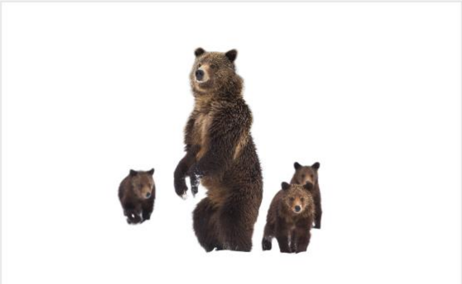
Original image	With background removed	Alpha matte
		
		
		

Image Analysis: Custom models for classification and object detection

- Select images that vary by: camera angle, lighting, background, visual style, individual/grouped subject(s), size, type.
- JPEG, PNG, GIF, BMP, WEBP, ICO, TIFF, or MPO format.
- COCO file


```
"images": [
  {
    "id": 1,
    "width": 500,
    "height": 828,
    "file_name": "0.jpg",
    "absolute_url": "https://blobstorage1.blob.core.windows.net/cpgcontainer/0.jpg"
  },
   {
     "id": 2,
     "width": 754,
     "height": 832,
     "file_name": "1.jpg",
     "absolute_url": "https://blobstorage1.blob.core.windows.net/cpgcontainer/1.jpg"
   },
```

```
"annotations": [
  {
    "id": 1,
    "category_id": 7,
    "image_id": 1,
    "area": 0.407,
    "bbox": [
      0.02663142641129032,
      0.40691584277841153,
      0.9524163571731749,
      0.42766634515266866
    ]
  },
  {
    "id": 2,
    "category_id": 9,
    "image_id": 2,
    "area": 0.27,
    "bbox": [
      0.11803319477782331,
      0.41586723392402375,
      0.7765206955096307,
      0.3483334397217212
    ]
  },
  ...

],
```

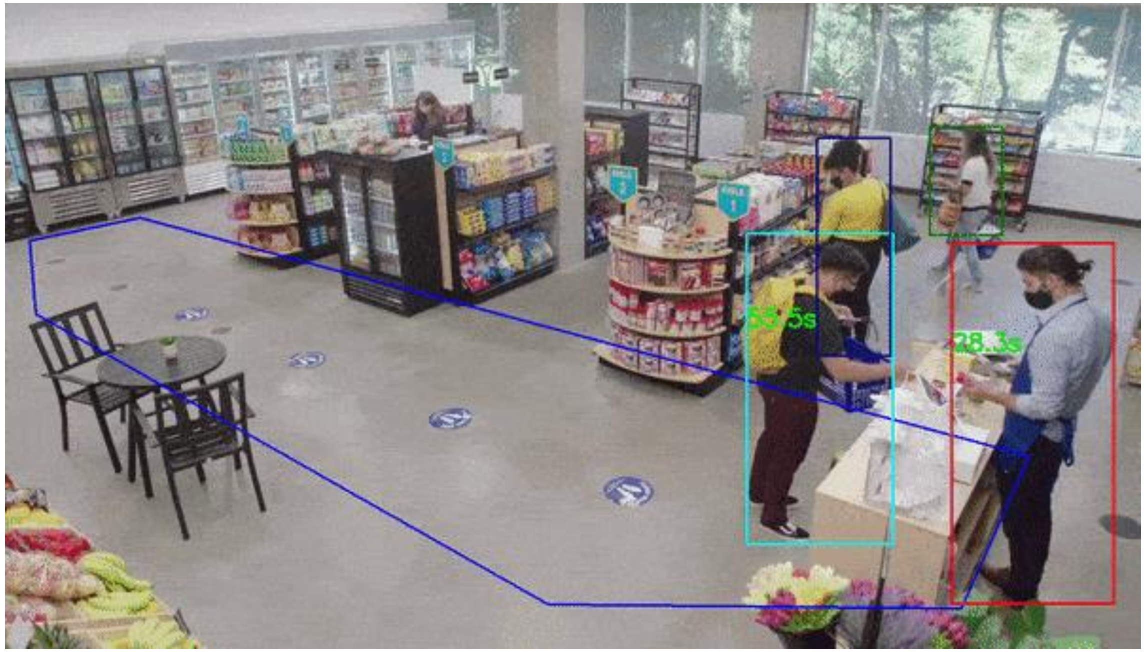
```
"categories": [
  {
    "id": 1,
    "name": "vegall original mixed vegetables"
  },
  {
    "id": 2,
    "name": "Amy's organic soups lentil vegetable"
  },
  {
    "id": 3,
    "name": "Arrowhead 8oz"
  },

  ...

]
```

Video Analysis

- Person counting
- Entrance Counting
  - wait times for a checkout line
  - measure foot traffic in a lobby or a specific floor



Video Analysis

- Social distancing and face mask detection

- Video Retrieval
  - Create a search index
  - Add documents (videos and images) to it
  - Search with natural language.

Azure AI Language

Extract information

Use Natural Language Understanding (NLU) to extract information from unstructured text. For example, identify key phrases or Personally Identifiable Information (PII), summarize text, recognize and categorize named entities, or customize an entity extraction model on top of your...

- Extract key phrases
- Find linked entities
- Named Entity Recognition (NER)
- Personally Identifiable Information (PII) detection
- Custom Named Entity Recognition (custom NER)
- Text analytics for health
- Custom text analytics for health

Summarize text-based content

Summarize lengthy documents and conversation transcripts.

- Document summarization
- Conversation summarization

Classify Text

Use Natural Language Understanding (NLU) to detect the language or classify the sentiment of text you have. You can also classify your text documents by customizing a classification model over your dataset.

- Analyze sentiment and mine text for opinions
- Detect Language
- Custom text classification

Answer questions

Provide answers to questions being asked in unstructured texts using our prebuilt capabilities, or customize your domain specific question-and-answer pairs over data you provide.

- Question answering

Understand conversations

Create your own models to classify conversational utterances and extract detailed information from them to fulfill scenarios.

- Conversational language understanding
- Orchestration workflow

Translate text

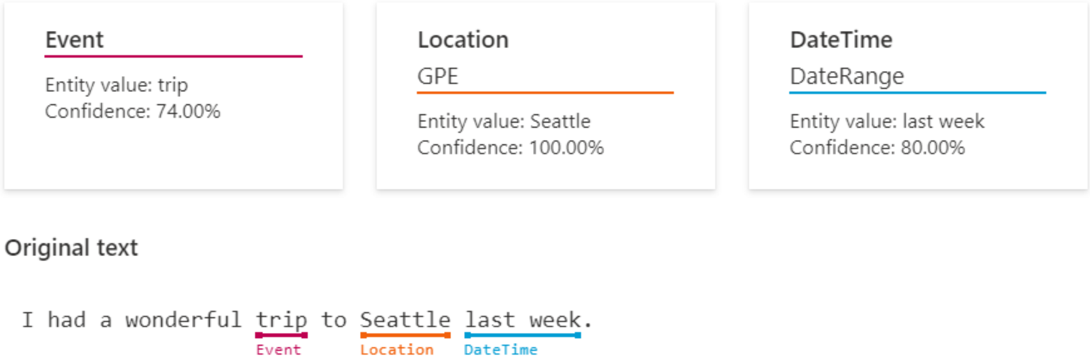
Use cloud-based neural machine translation to build intelligent, multi-language solutions for your applications.

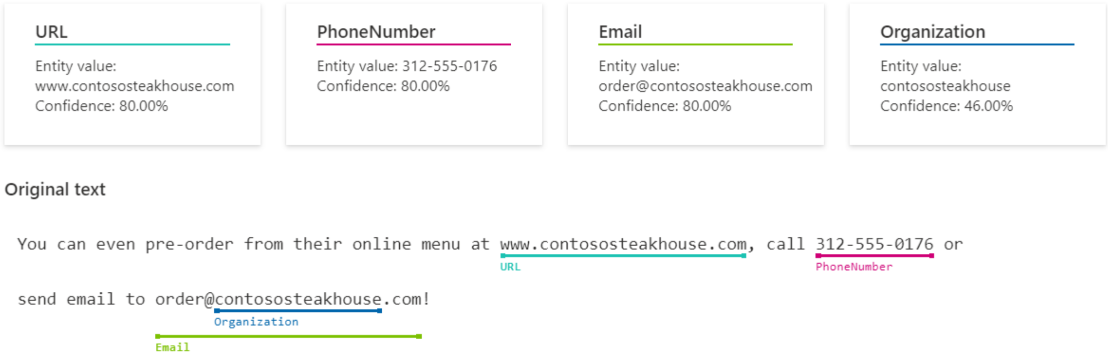
- Use machine translation on text.

Azure AI Language Features

- Named Entity Recognition (NER)
  - Categorizes entities (words or phrases) in unstructured text across several predefined category groups. For example: people, events, places, dates, and more.

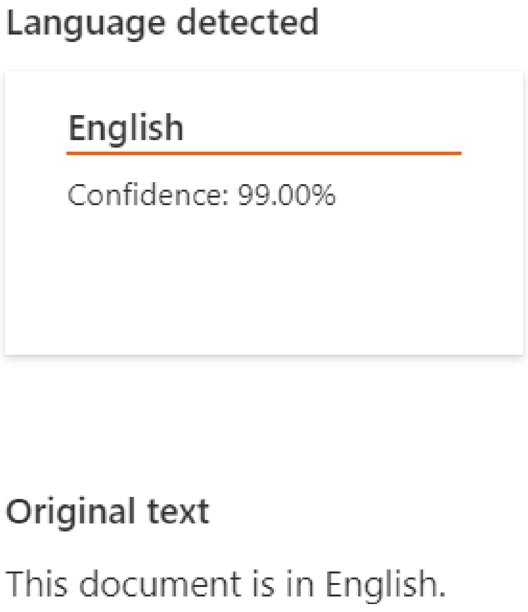
- Personally identifying (PII) and health (PHI) information detection
  - Identifies, categorizes, and redacts sensitive information in both unstructured text documents, and conversation transcripts. For example: phone numbers, email addresses, forms of identification, and more.
  - Customers can now redact transcripts, chats, and other text written in a conversational style (i.e. text with “um”s, “ah”s, multiple speakers etc.)



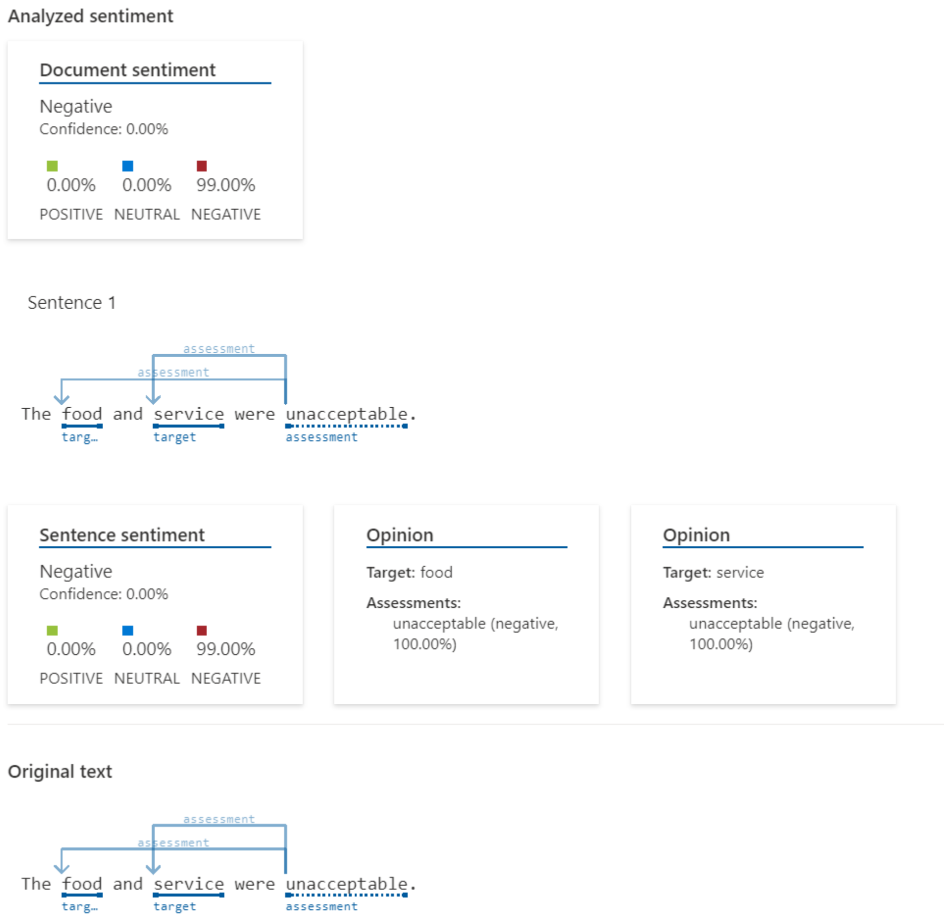


Azure AI Language Features

- Language and script detection
  - Detect the language a document is written in, and returns a language code for a wide range of languages, variants, dialects, and some regional/cultural languages.



- Sentiment Analysis and opinion mining



Azure AI Language Features

- Summarization: Uses extractive text summarization to produce a summary of documents and conversation transcriptions. It extracts sentences that collectively represent the most important or relevant information within the original content.

Sentence 1

We're delighted to announce that Cognitive Service for Language service now supports extractive summarization!

Rank Score

100.00%

Sentence 2

Document summarization is a feature that produces a text summary by extracting sentences that collectively represent the most important or relevant information within the original content.

Rank Score

39.00%

Sentence 3

Extractive summarization condenses articles, papers, or documents to key sentences.

Rank Score

96.00%

Original text

We're delighted to announce that Cognitive Service for Language service now supports extractive summarization! In general, there are two approaches for automatic document summarization: extractive and abstractive. This feature provides extractive summarization. Document summarization is a feature that produces a text summary by extracting sentences that collectively represent the most important or relevant information within the original content. This feature is designed to shorten content that could be considered too long to read. Extractive summarization condenses articles, papers, or documents to key sentences.

Rank Score: 100.00%

Rank Score: 39.00%

Rank Score: 96.00%

- Key phrase extraction
  - Evaluates and returns the main concepts in unstructured text, and returns them as a list.

Key phrases

Pike place market, favorite Seattle attraction

Original text

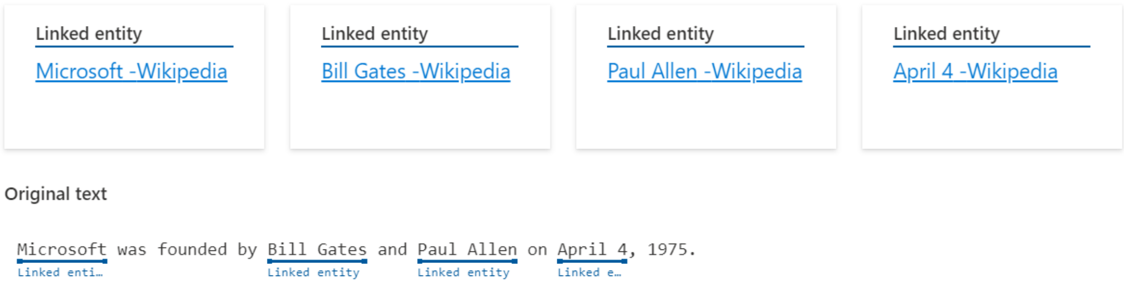
Pike place market is my favorite Seattle attraction.

Key phrase

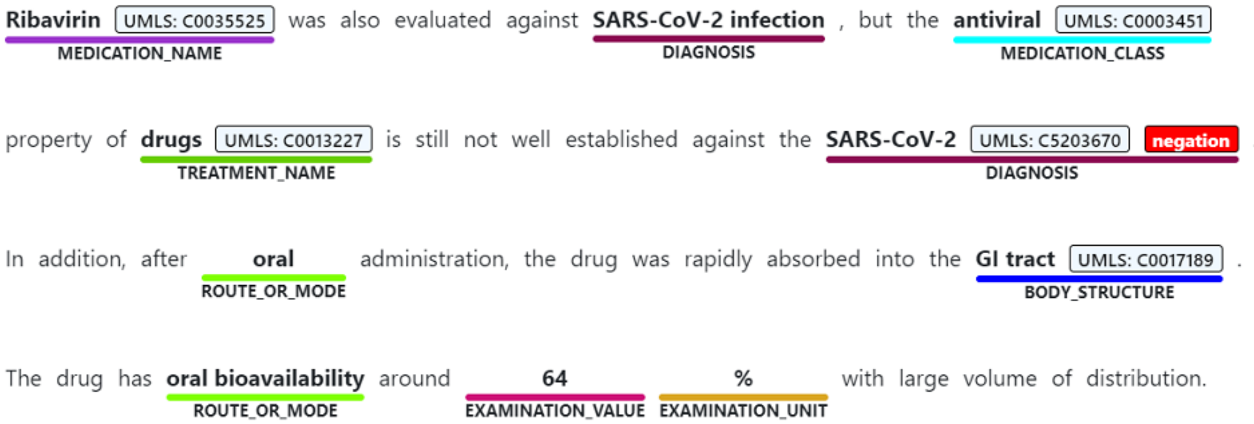
Key phrase

Azure AI Language Features

- Entity linking
  - Disambiguates the identity of entities (words or phrases) found in unstructured text and returns links to Wikipedia.

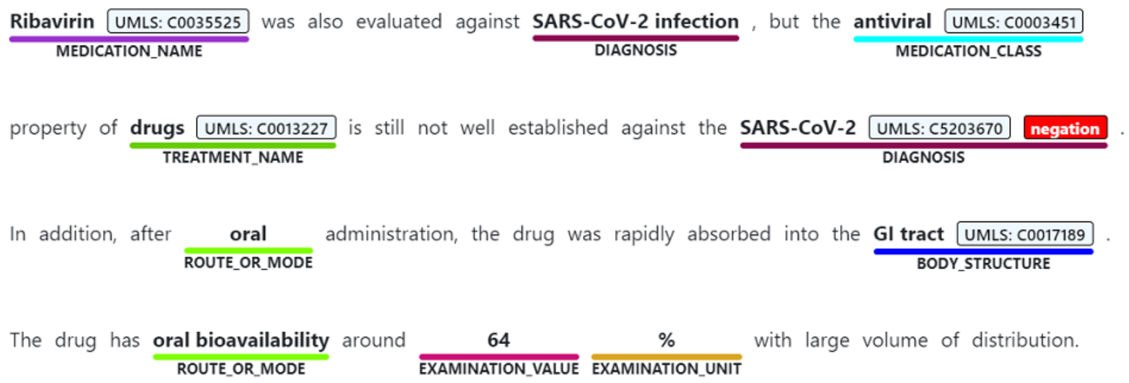


- Text analytics for health
  - Extracts and labels relevant medical information from unstructured texts such as doctor's notes, discharge summaries, clinical documents, and electronic health records.



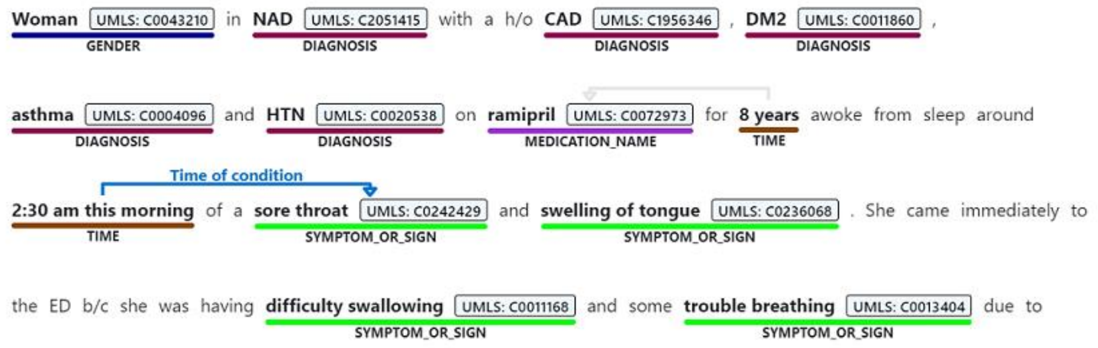
Text analytics for health

Named entity recognition



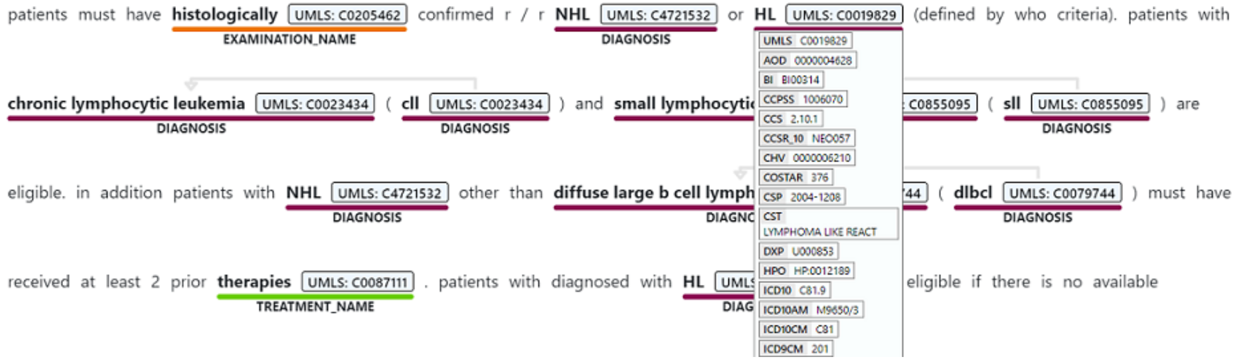
[Supported entity types.]

Relation extraction

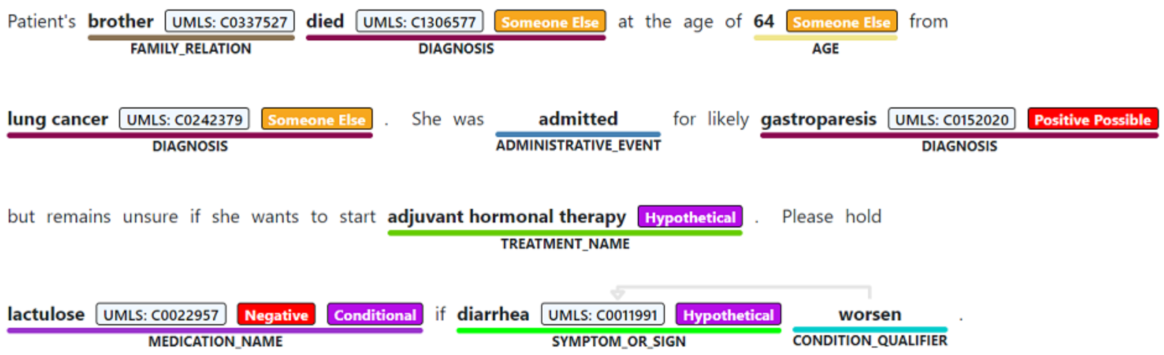


[Supported relation types.]

Entity linking (to UMLS – Unified Medical Lang System)



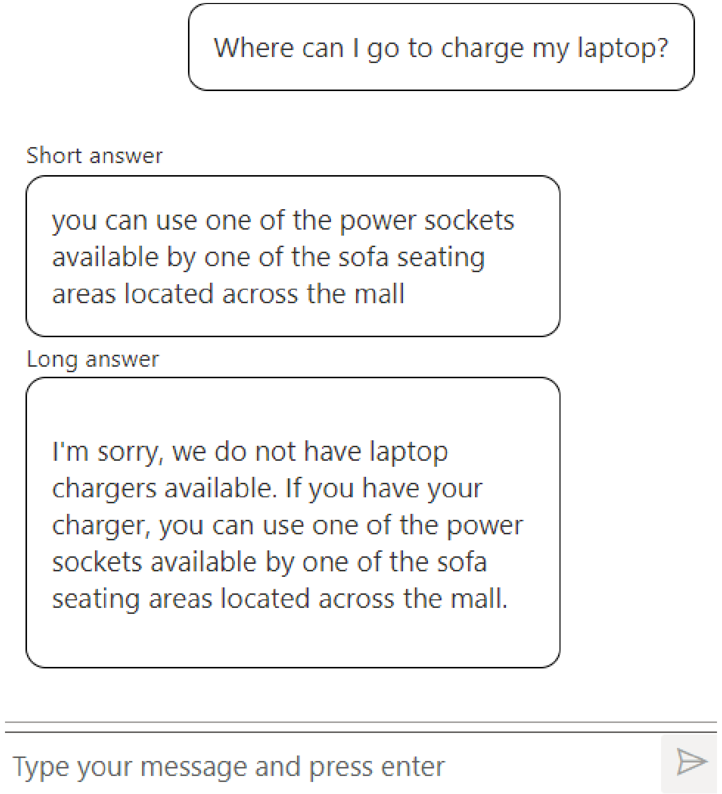
Assertion detection



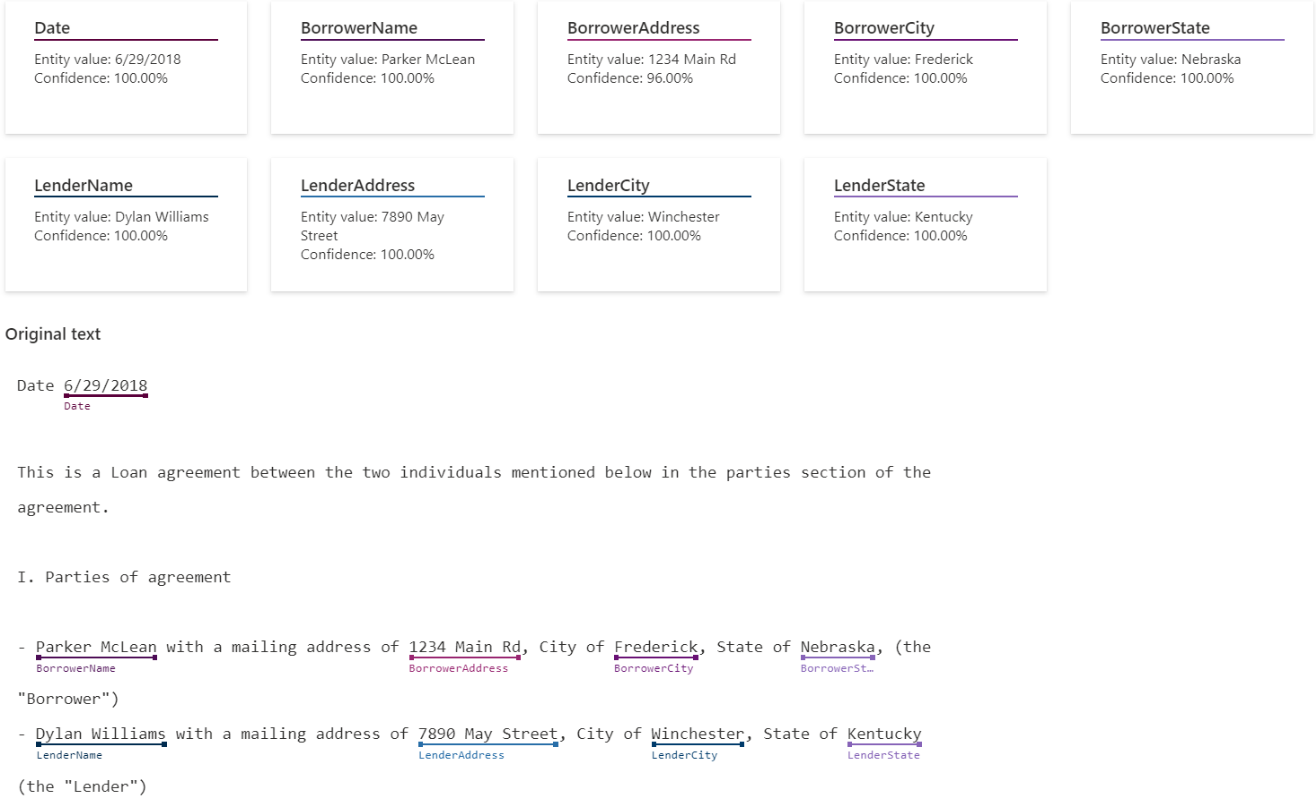
[Supported assertions]

Azure AI Language Features

- Question answering
  - Finds the most appropriate answer for inputs from your users, and is commonly used to build conversational client applications, such as social media applications, chat bots, and speech-enabled desktop applications.
  - QnA Maker

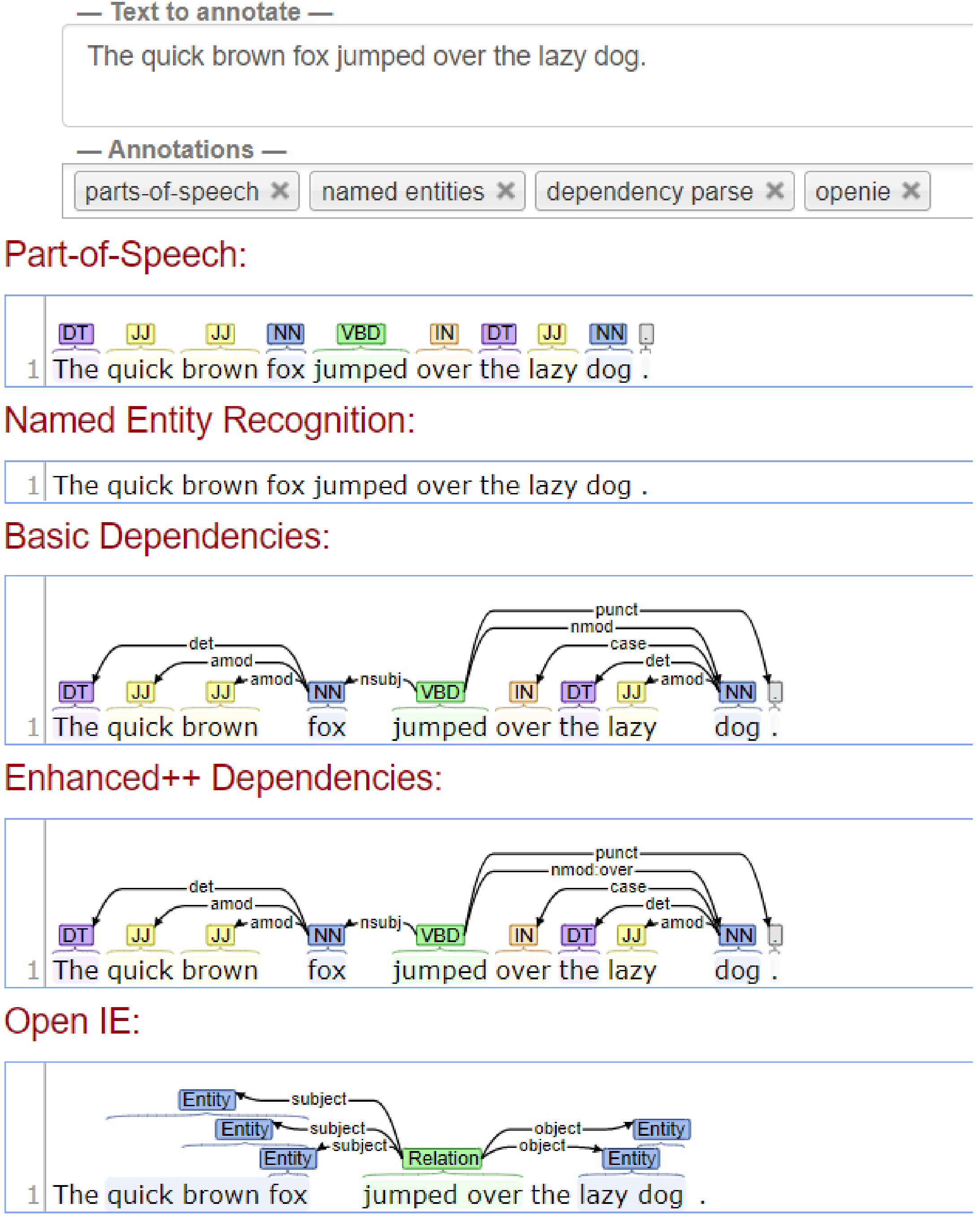


- Custom Named Entity Recognition (Custom NER)
  - Build custom AI models to extract custom entity categories (labels for words or phrases), using unstructured text that you provide.

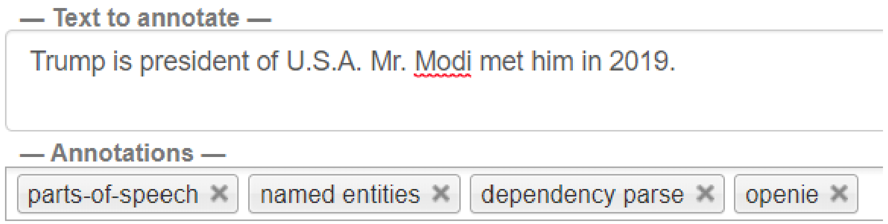


http://corenlp.run/

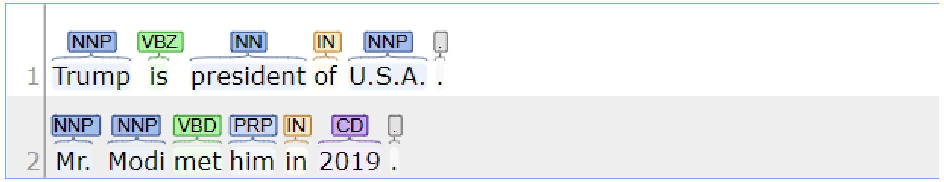
https://tagme.d4science.org/tagme/



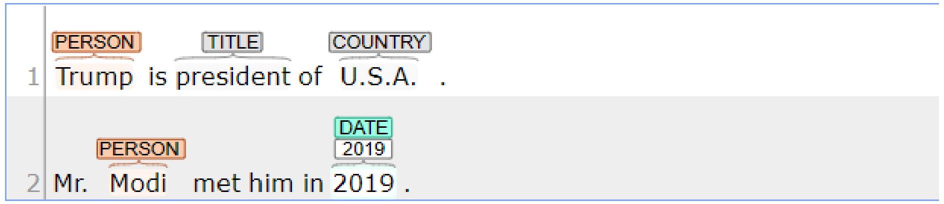
http://corenlp.run/



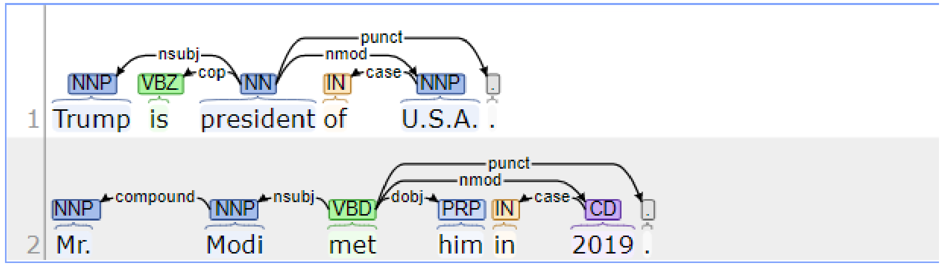
Part-of-Speech:



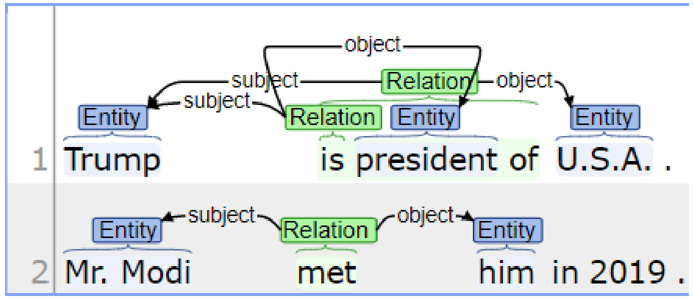
Named Entity Recognition:



Basic Dependencies:



Open IE:



Content Moderation

Image moderation

Enhance your ability to detect potentially offensive or unwanted images through machine learning-based classifiers, custom lists and optical character recognition (OCR).

Text moderation

Use content filtering to detect potential profanity in more than 100 languages, flag text that may be deemed inappropriate depending on context (in public preview) and match text against your custom lists. Content Moderator also helps check for personally identifiable information (PII).

Video moderation

Enable machine-assisted detection of possible adult and racy content in videos. The video moderation service (in public preview) is available as part of Azure Media Services.

Human review tool

The best content moderation results come from humans and machines working together. Use the review tool when prediction confidence can be improved or tempered with a real-world context.

Azure AI Document Intelligence

- Quickly extract text, key-value pairs, tables and structure from documents

Simple text extraction

Easily pull data and organize information with prebuilt and custom features—no manual labeling required.

Customized results

Get output tailored to your layouts with automatic custom extraction and improve it with human feedback.

Flexible deployment

Ingest data from the cloud or at the edge and apply to search indexes, business automation workflows, and more.

Built-in security

Rely on enterprise-grade security and privacy applied to both your data and any trained models.

Analyze forms and documents

Make data-driven decisions by extracting data from documents and putting it into your data visualization service for analysis.

Create intelligent search indexes

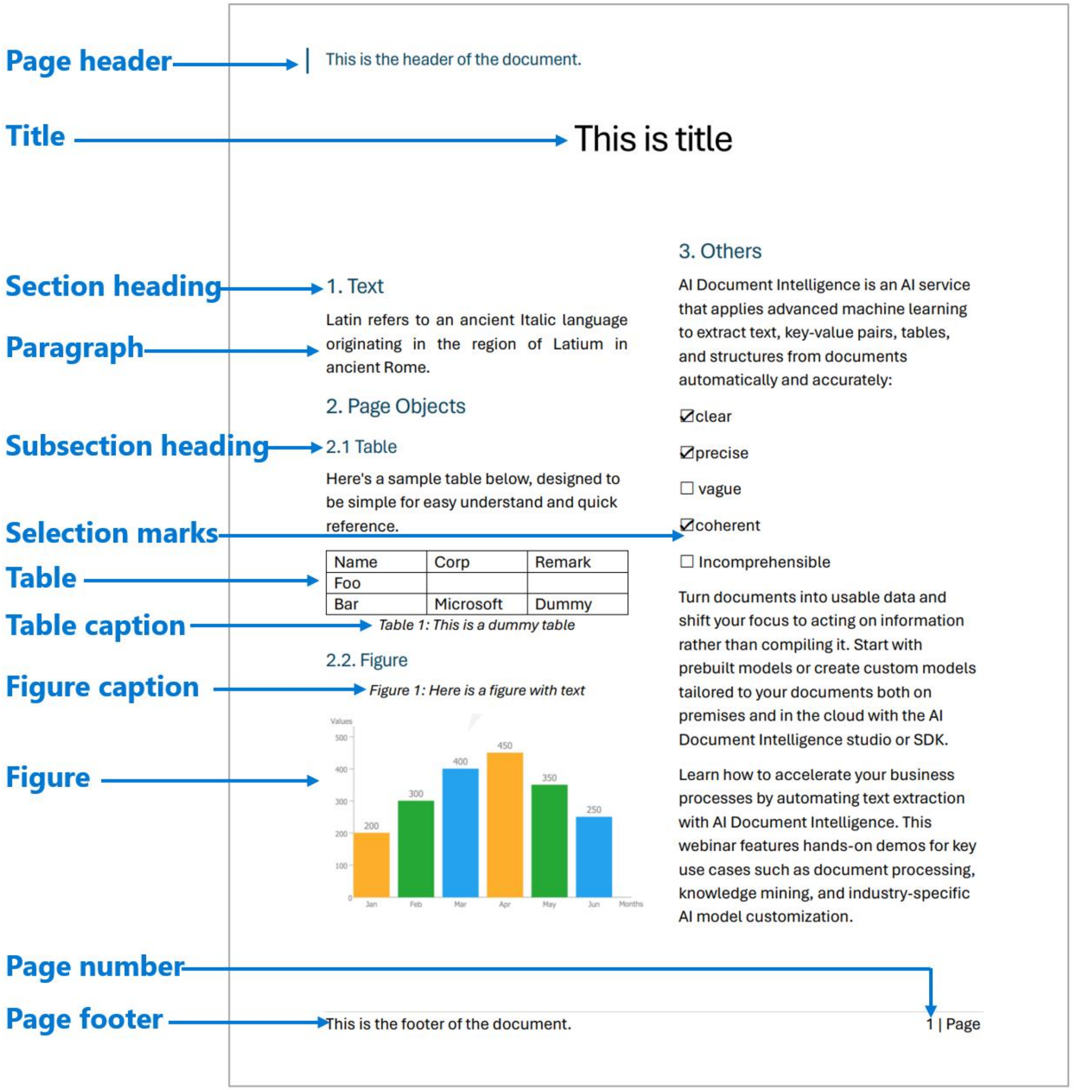
Easily find specific information in your documents and forms, such as total accounts payable, by integrating AI Document Intelligence with Azure Applied AI Search.

Automate business workflows

Extract text, key-value pairs, tables, and structures from documents, and pipe them into your back-end systems to perform tasks such as claim, invoice, and receipt processing.

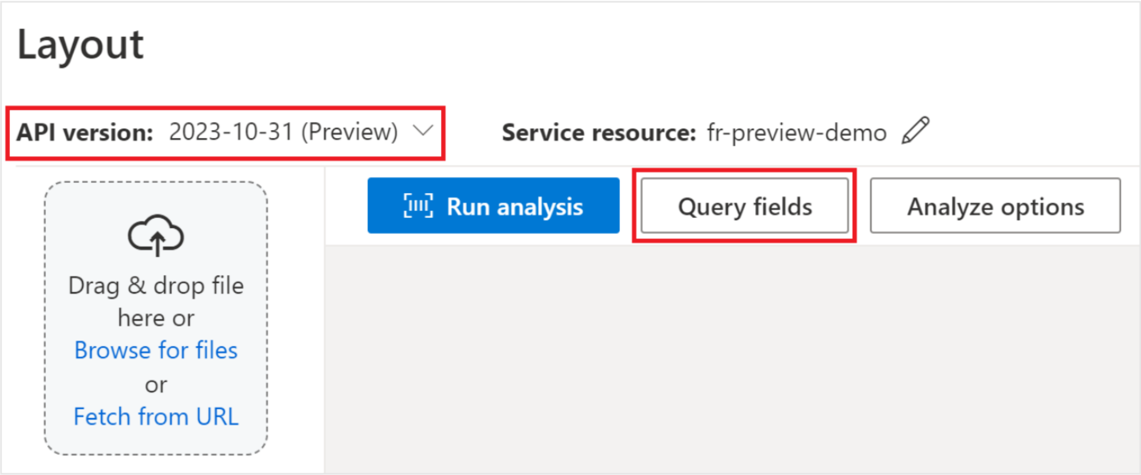
Document Intelligence read and layout model

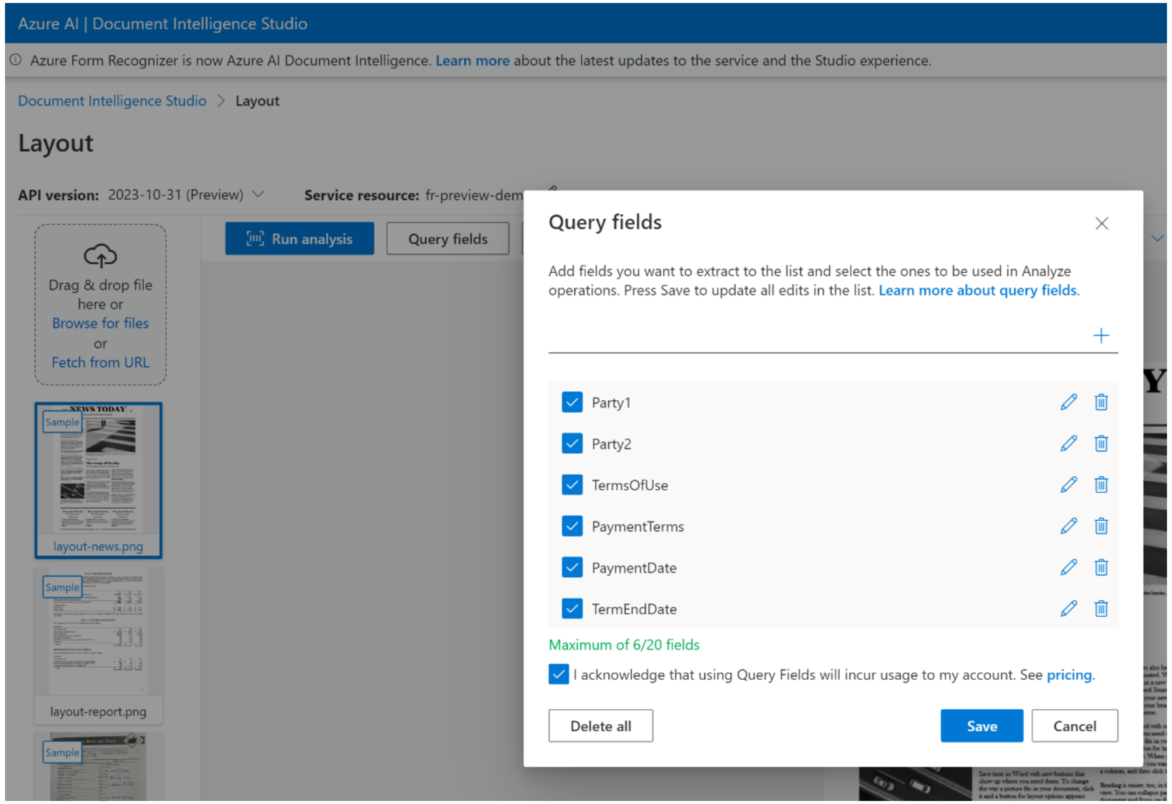
- Optical Character Recognition (OCR) for documents is optimized for large text-heavy documents in multiple file formats and global languages.
- Paragraph extraction
  - Text block, bounding polygon coordinates.
- Text, lines, and words extraction
  - Extracts print and handwritten style text as lines and words.
  - Bounding polygon coordinates and confidence for the extracted words.
  - Spans pointing to the associated text.
- Formulas, such as mathematical equations
- All font properties of text
  - similarFontFamily, fontStyle for styles such as italic and normal, fontWeight for bold or normal, color for color of the text, and backgroundColor for color of the text bounding box.
- Barcode property extraction
- Language detection
- Convert an analog PDF, such as scanned-image PDF files, to a PDF with embedded text.



Query Field Extraction

- specify the fields you want to extract and Document Intelligence analyzes the document accordingly.
- You can pass a list of field labels like Party1, Party2, TermsOfUse, PaymentTerms, PaymentDate, and TermEndDate as part of the analyze document request.





Prebuilt Doc Intelligence Extraction Models

Financial Services and Legal

Bank Statement | Extract account information and details from bank statements.

Check | Extract relevant information from checks.

Contract | Extract agreement and party details.

Credit card | Extract payment card information.

Invoice | Extract customer and vendor details.

Pay Stub | Extract pay stub details.

Receipt | Extract sales transaction details.

US Tax

Unified US tax | Extract from any US tax forms supported.

US Tax W-2 | Extract taxable compensation details.

US Tax 1098 | Extract `1098` variation details.

US Tax 1099 | Extract `1099` variation details.

US Tax 1040 | Extract `1040` variation details.

US Mortgage

US mortgage 1003 | Extract loan application details.

US mortgage 1004 | Extract information from appraisal.

US mortgage 1005 | Extract information from validation of employment.

US mortgage 1008 | Extract loan transmittal details.

US mortgage disclosure | Extract final closing loan terms.

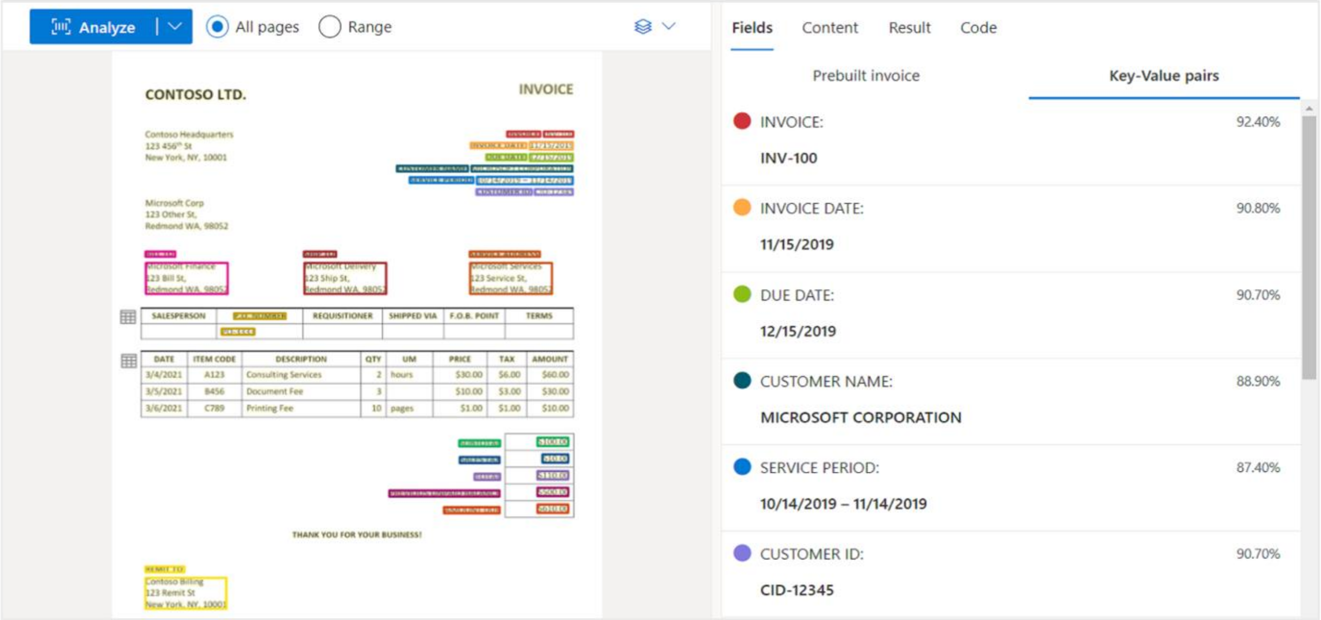
Personal Identification

Health Insurance card | Extract insurance coverage details.

Identity | Extract verification details.

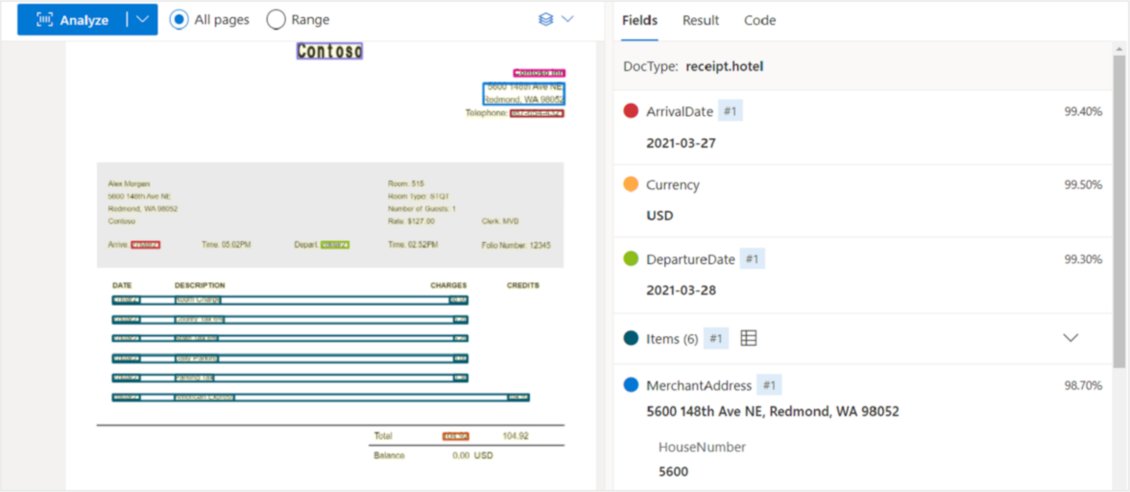
Marriage certificate | Extract certified marriage information.

Invoice

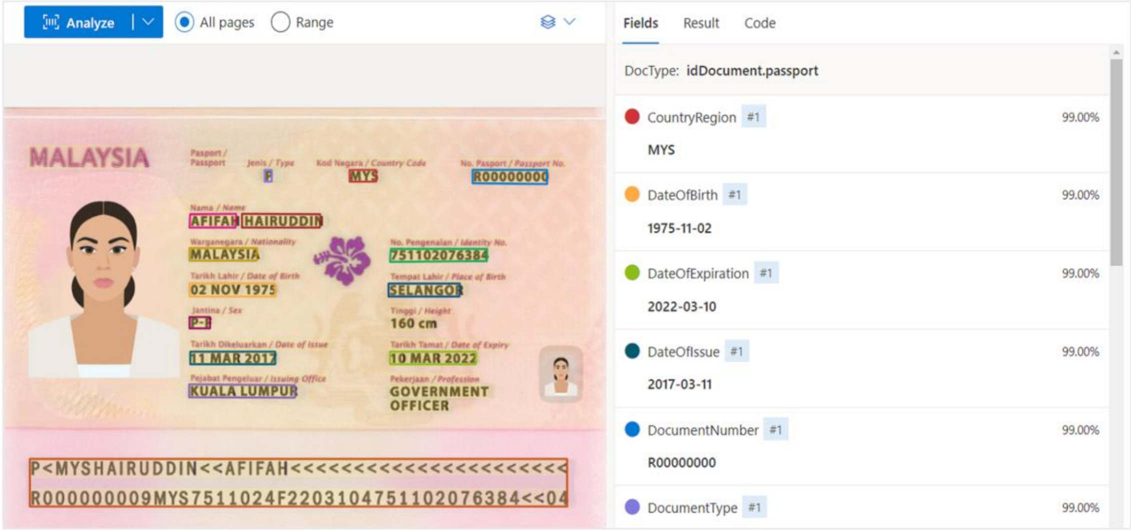


Prebuilt Doc Intelligence Extraction Models

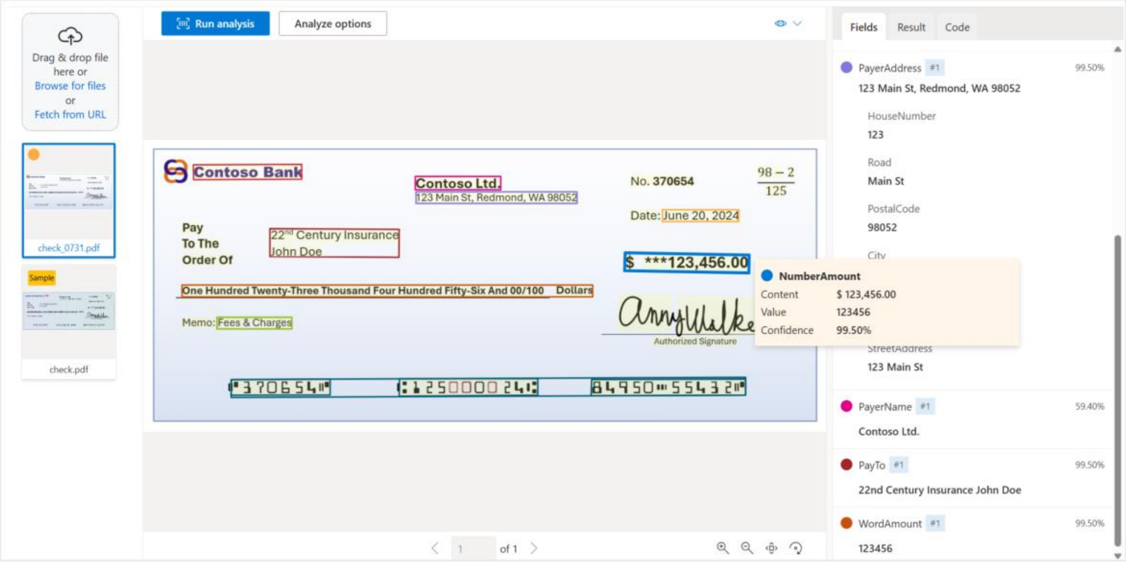
Receipt



Identity (ID)

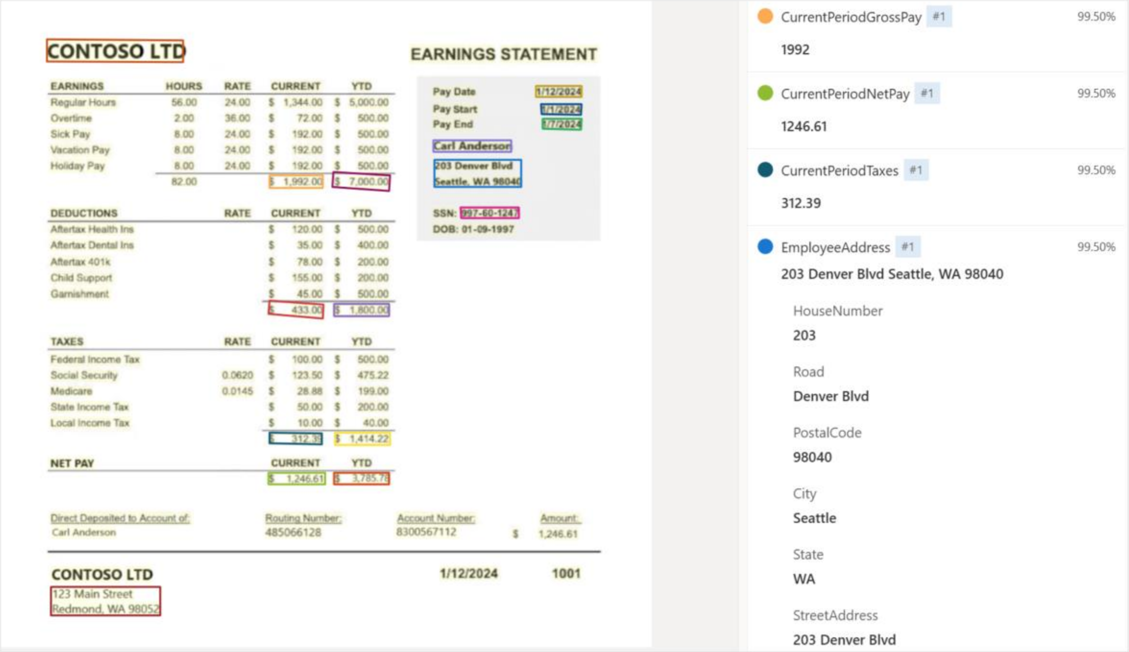


Check

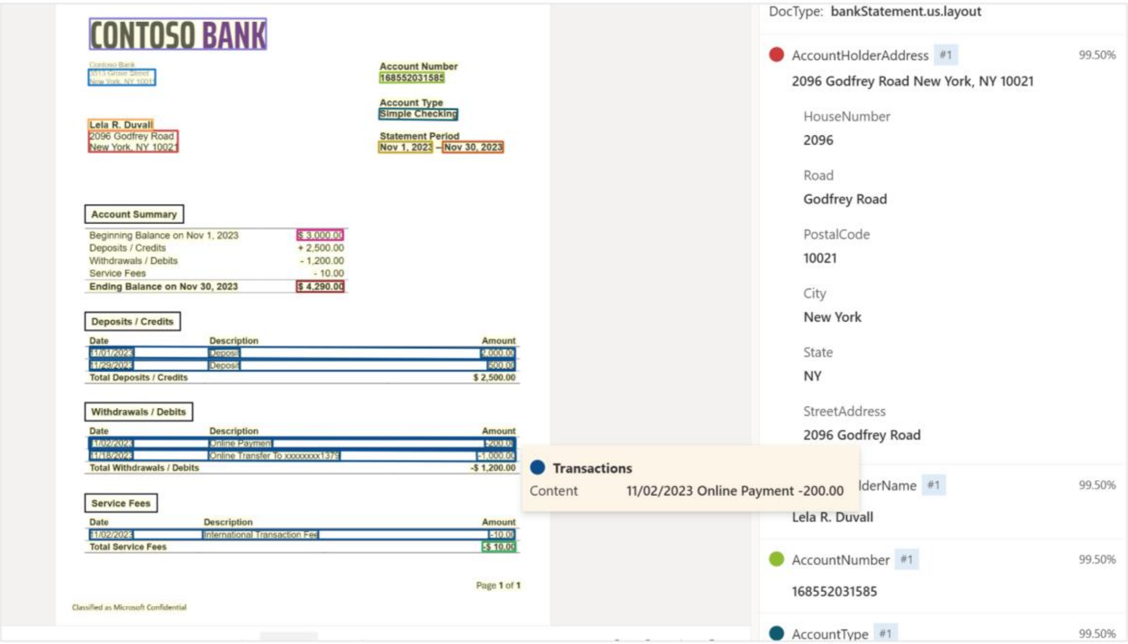


Prebuilt Doc Intelligence Extraction Models

Pay stub

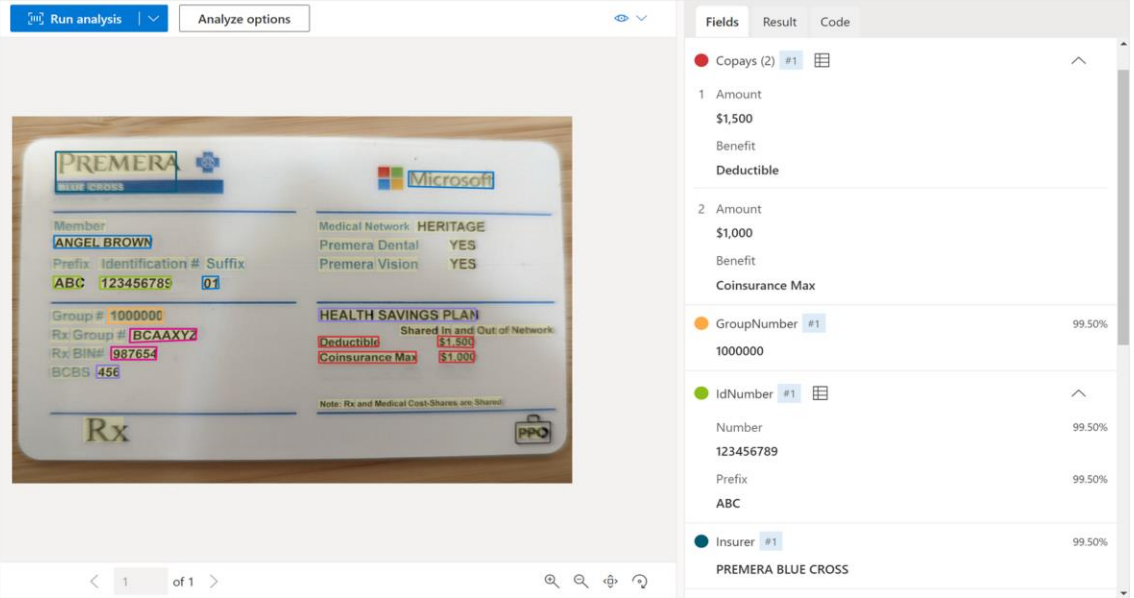


Bank statement

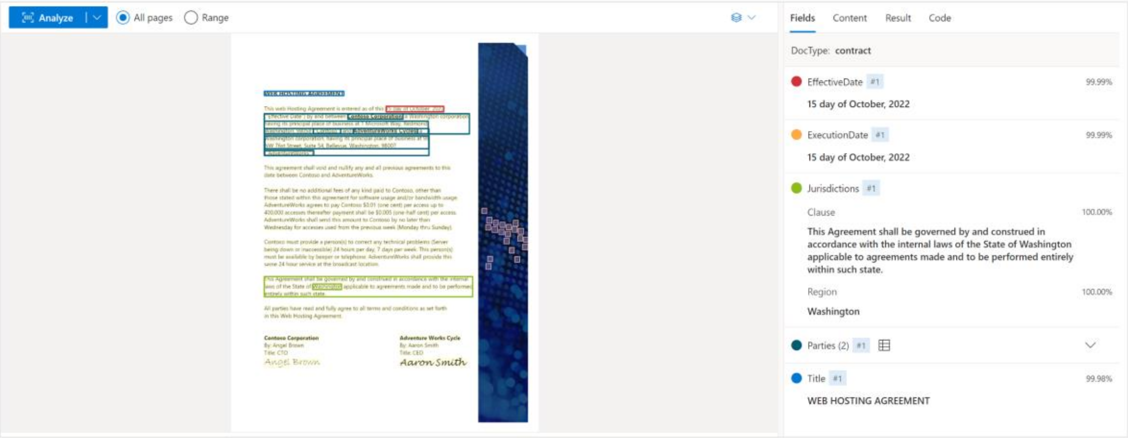


Prebuilt Doc Intelligence Extraction Models

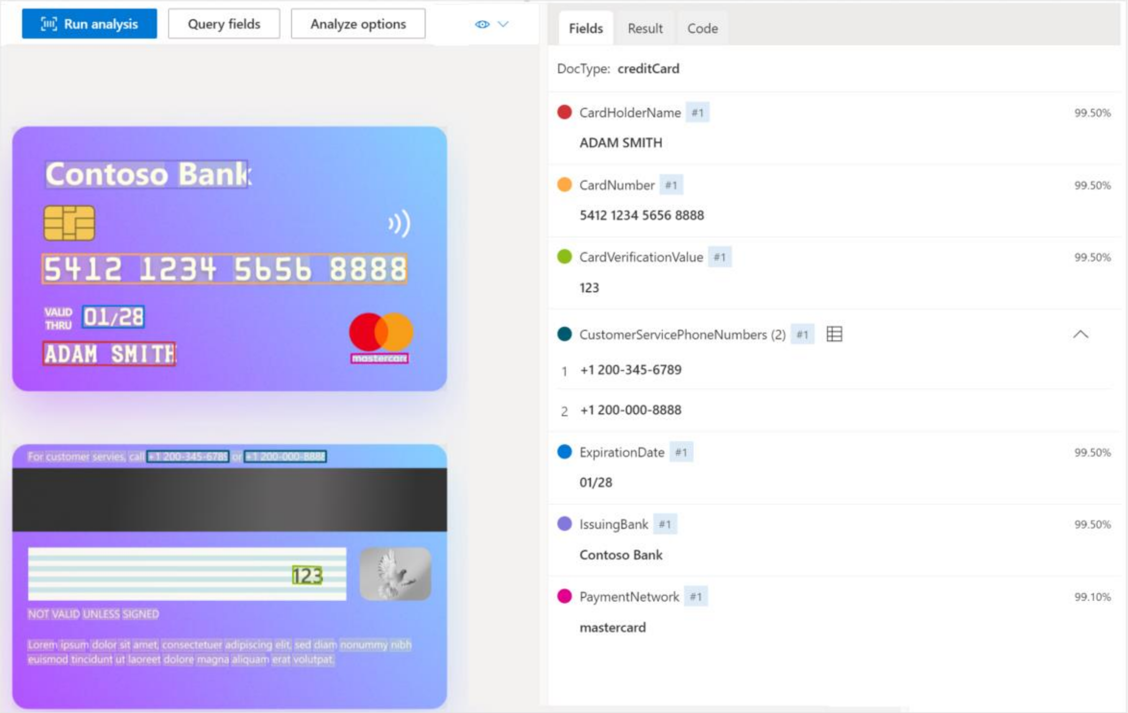
Health insurance card



Contract model



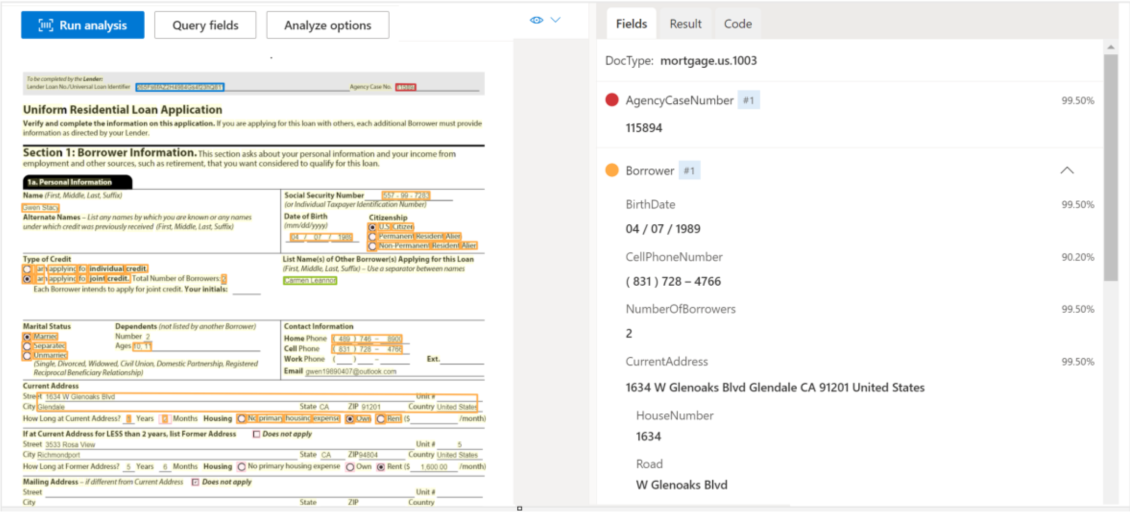
Credit card model



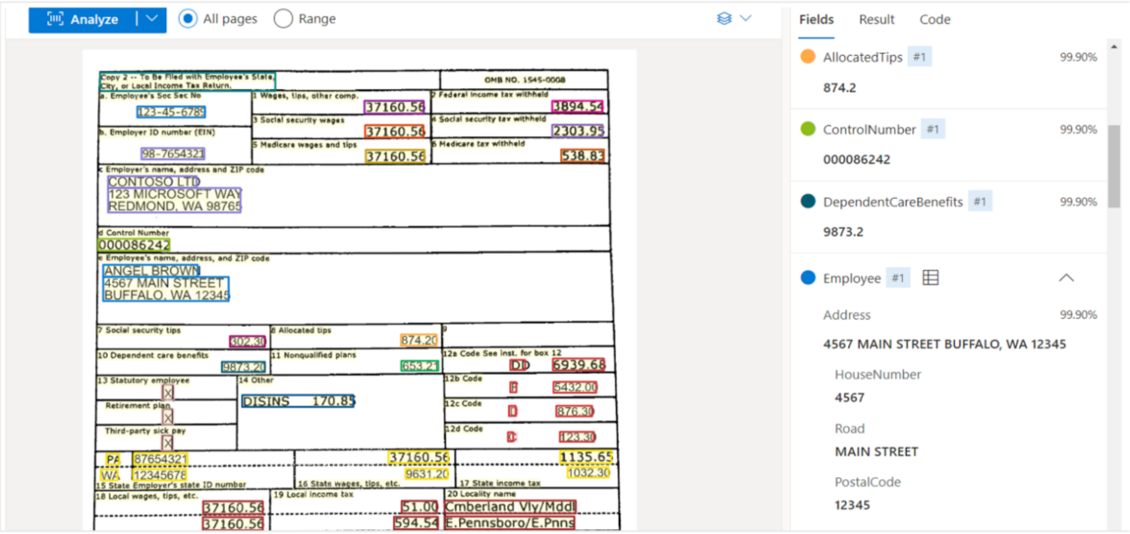
Prebuilt Doc Intelligence Extraction Models

Marriage certificate model

US mortgage 1003 form



US Tax W-2 model



Azure AI Speech

Transcribe speech to text

Transcribe call center or meeting conversations. Go global with audio-captioning in more than 100 languages.

Learn more

Convert text to speech

Build bots that speak naturally. Differentiate your brand with customized, realistic voices and speaking styles.

Learn more

Speech analytics

Analyze audio or video call recordings to gain deep insights. Summarize key topics and extract or redact personal identification information.

Learn more

Transcribe audio with OpenAI Whisper

Transform your call centers using the latest OpenAI Whisper model in Azure AI Speech or Azure OpenAI Service.

Read the blog

Build custom voices

Build natural-sounding voices with custom neural voice.

Learn more

Build your avatars

Bring your brand to life using pre-built or custom avatars with natural-sounding voices.

Learn more

Verify and recognize speakers

Confirm a person's identity or recognize who's speaking in a meeting by adding speaker verification and identification to your app.

Learn more

Enable multilingual communication

Translate audio or video data from and into an ever-growing list of supported languages. Customize translations to your industry.

Learn more

Embedd speech

Use embedded speech to power on-device speech to text and text to speech scenarios where cloud connectivity is intermittent or unavailable.

Learn more

Speech to text

- Applications
  - Transcriptions, captions, or subtitles for live meetings: Real-time audio transcription for accessibility and record-keeping.
  - Diarization: Identifying and distinguishing between different speakers in the audio.
  - Pronunciation assessment: Evaluating and providing feedback on pronunciation accuracy.
  - Call center agents assist: Providing real-time transcription to assist customer service representatives.
  - Dictation: Transcribing spoken words into written text for documentation purposes.
  - Voice agents: Enabling interactive voice response systems to transcribe user queries and commands.
- Caption and speech synchronization: synchronize captions with the audio track, whether it's in real-time or with a prerecording.
- Get partial results: Speech recognition results are subject to change while an utterance is still being recognized.
- Language identification: identify languages spoken in audio (once or continuous)
- Multilingual speech to speech.
- The Speech service supports output formats such as SRT (SubRip Text) and WebVTT (Web Video Text Tracks)

```
00:00:00,180 --> 00:00:03,230
Welcome to applied Mathematics course 201.
```

Text to Speech

- Real-time speech synthesis
- Asynchronous synthesis of long audio
- Speech Synthesis Markup Language (SSML)
  - Used to customize text to speech outputs.
  - Adjust pitch, add pauses, improve pronunciation, change speaking rate, adjust volume, and attribute multiple voices to a single document.
- Visemes
  - Key poses in observed speech, including the position of the lips, jaw, and tongue in producing a particular phoneme.
  - Visemes can be used to generate facial animation data
- Text to speech avatar (speech as well as video)

Gender	Female, Male, Neutral
Age group	Child, Young adult, Adult, Senior
Capability	Context-aware, Multi-lingual, Multi-style
Recommendation	Featured, Popular
Speaking style	Advertisement - upbeat, Affectionate, Angry, Assistant, Calm, Chat, Chat - casual, Cheerful, Conversation, Customer serv, More
Personality	Animated, Approachable, Authentic, Authoritative, Bright, Calm, Caring, Casual, Cheerful, Clear, Confident, Cris, More
Tailored scenario	Advertisement, Assistant, Audiobook, Audiobooks, Chat, Documentary, E-learning, Gaming, Meditation, Narration, More

Pronunciation assessment

Parameter	Description	Granularity
AccuracyScore	How closely the phonemes match a native speaker's pronunciation. Syllable, word, and full text accuracy scores are aggregated from the phoneme-level accuracy score, and refined with assessment objectives.	Phoneme level, Syllable level (en-US only), Word level, Full Text level
FluencyScore	How closely the speech matches a native speaker's use of silent breaks between words.	Full Text level
CompletenessScore	Completeness of the speech, calculated by the ratio of pronounced words to the input reference text.	Full Text level
ProsodyScore	How natural the given speech is, including stress, intonation, speaking speed, and rhythm.	Full Text level
PronScore	Overall score of the pronunciation quality of the given speech, calculated from AccuracyScore, FluencyScore, CompletenessScore, and ProsodyScore with weight	Full Text level
ErrorType	• Whether a word is omitted, inserted, or improperly inserted with a break. • Indicates a missing break at punctuation. • Indicates whether a word is badly pronounced, or monotonically rising, falling, or flat on the utterance. Mispronunciation when the pronunciation AccuracyScore for a word is below 60.	Word level
GrammarScore	• Correctness in using grammar and variety of sentence patterns. Lexical accuracy, grammatical accuracy, and diversity of sentence structures jointly elevate grammatical errors.	Full Text level
TopicScore	• Level of understanding and engagement with the topic	Full Text level

Azure OpenAI service

- Provides REST API access to OpenAI's powerful language models including o1-preview, o1-mini, GPT-4o, GPT-4o mini, GPT-4 Turbo with Vision, GPT-4, GPT-3.5-Turbo, and Embeddings model series.
- Models available
  - o1-preview & o1-mini
  - GPT-4o & GPT-4o mini
  - GPT-4o audio
  - GPT-4 series (including GPT-4 Turbo with Vision)
  - GPT-3.5-Turbo series
  - Embeddings series (text-embedding-3-large, text-embedding-3-small, text-embedding-ada-002)
  - DALLE, Whisper.
- Fine-tuning
  - GPT-4o-mini (preview)
  - GPT-4 (preview)
  - GPT-3.5-Turbo (0613)
  - babbage-002
  - davinci-002.

Model ID	Max request (tokens)	Training Data (up to)
babbage-002	16,384	Sep 2021
davinci-002	16,384	Sep 2021
gpt-35-turbo (0613)	4,096	Sep 2021
gpt-35-turbo (1106)	Input: 16,385 Output: 4,096	Sep 2021
gpt-35-turbo (0125)	16,385	Sep 2021
gpt-4 (0613) [1]	8192	Sep 2021
gpt-4o-mini [1] (2024-07-18)	Input: 128,000 Output: 16,384 Training example context length: 64,536	Oct 2023
gpt-4o [1] (2024-08-06)	Input: 128,000 Output: 16,384 Training example context length: 64,536	Oct 2023

Azure AI Content Safety

Safety system for monitoring content generated by both LLMs and humans.

Moderate text content

Run moderation tests on text contents. Assess the test results with detected severities. Experiment with different threshold levels.

Groundedness detection

Groundedness detection detects ungroundedness generated by the large language models (LLMs).

Protected material detection (Text)

Use protected material detection to detect and protect third-party text material in LLM output.

Prompt Shields

Prompt Shields provides a unified API that addresses Jailbreak attacks and Indirect attacks.

Moderate image content

Run moderation tests on image contents. Assess the test results with detected severities. Experiment with different threshold levels.

Moderate multimodal content

Run moderation tests on image and text combined contents. Assess the test results with detected severities.

Custom categories

Create and manage content categories specific to your needs for enhanced moderation and filtering.

Safety system message

Use the framework of metaprompt that helps you potentially mitigate different types of harm.

Search APIs

Bing Autosuggest

Help users complete queries faster by adding intelligent type-ahead capabilities

Bing Custom Search

Create a custom search engine with ad-free results.

Bing Entity Search

Recognize and classify named entities and find search results based on them.

Bing Image Search

Add a variety of search image options to your app.

Bing News Search

Turn any app into a news search resource.

Bing Spell Check

Help users identify and fix word breaks, slang, names, homonyms and brands.

Bing Video Search

Add advanced video search features to your app.

Bing Visual Search

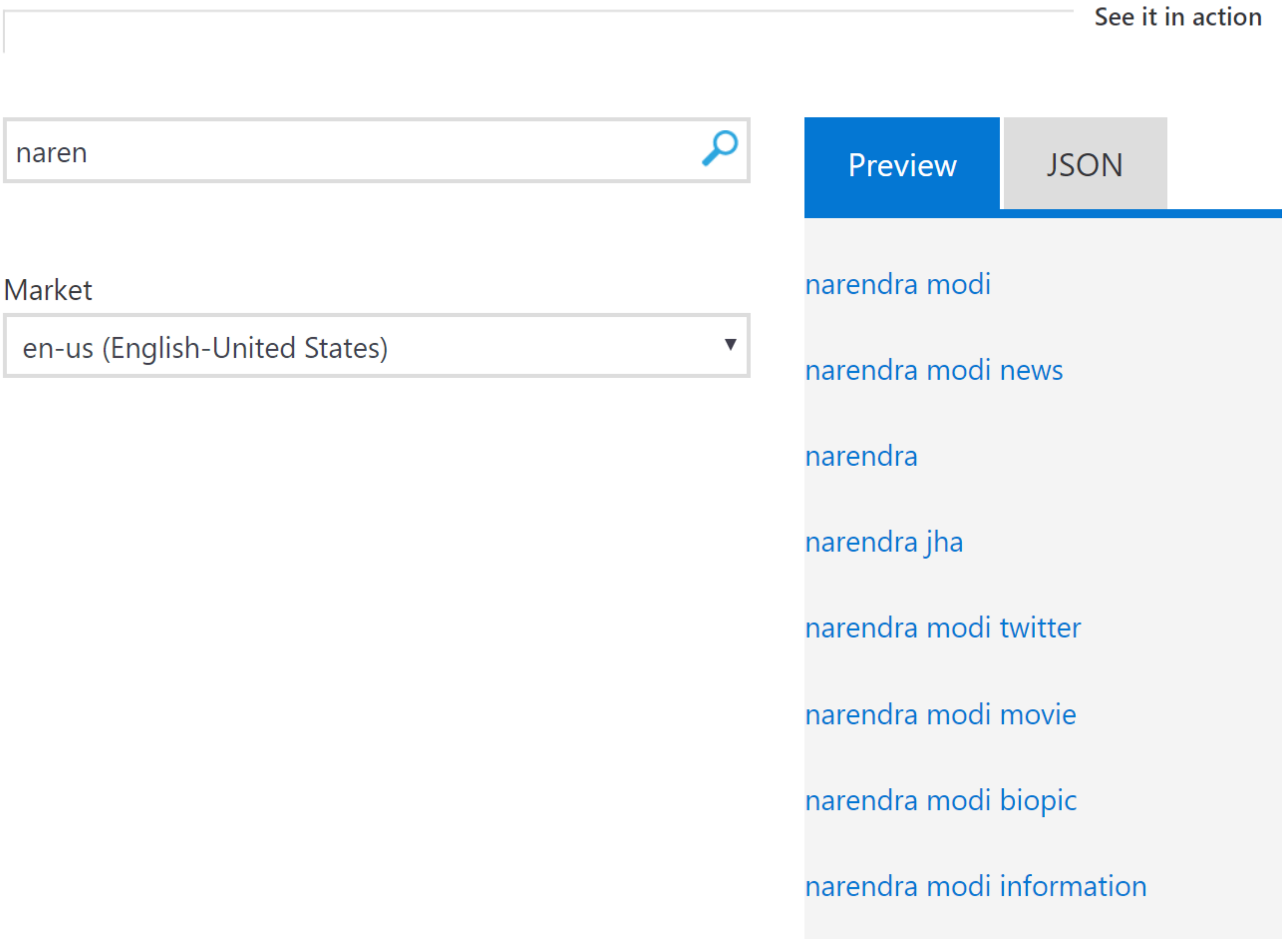
Enable users to search using images.

Bing Web Search

Enable safe, ad-free, location-aware search for your users, surfacing relevant information from web results, images, news, videos and visuals.

AutoSuggest

https://www.microsoft.com/en-us/bing/apis/bing-autosuggest-api



Thanks!!

Questions?